

Tradelatest — Repository Encyclopedia (Grok)

Charter for remaining files (Groups A-D), ~403 NOT_IN_BOOK, path from architecture guide to full repository encyclopedia.

Generated 2026-08-07 from docs/book/
Grok review pass export — reading convenience only.
On conflict, repository code/config wins.

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Chapter 24 - Repository Encyclopedia

Part IX - Toward Full Repository Understanding

Status of this chapter: Written (charter + remaining-file map, Grok pass 2026-08-07)

Also known as: the Repository Encyclopedia plan - how the Canonical Knowledge Book (architecture guide) becomes a file-level encyclopedia without treating every unread file as equally important.

Why this chapter exists

The Canonical Knowledge Book (Chapters 00-23 + A1-A2) explains the **production spine and governance story**. It was never meant to name every Python file. After the coverage report (`grok/Book_PDF_File_Coverage_Grok.xlsx`), the natural question is:

What about the remaining files? Are they dormant? Unimportant? Or just not chaptered yet?

This chapter answers that. The remaining files fall into **several categories**. They are **not all equally important**, and the coverage report already hints at that.

What problem it solves

Gives a durable plan for “100% repository understanding” that:

1. Preserves the book’s architecture spine (do not rewrite it into a flat file dump).
2. Prioritizes **core omissions** over scripts and dormant packages.
3. Defines what a true **Repository Encyclopedia** entry looks like per significant file.

What you need to already know

- [Chapter 00](#) - candle -> order spine.
- [Chapter 04](#) - package map + un-chaptered orientation table.
- [Appendix A2](#) - open book gaps.
- Coverage spreadsheet: `grok/Book_PDF_File_Coverage_Grok.xlsx` (regenerate via `python grok/book_file_coverage_report.py`).

The idea

Current book vs encyclopedia

Mode	What it optimizes for	File coverage style
Canonical Knowledge Book (Ch.00-23)	Architecture, invariants, production path, research discipline	Concepts + load-bearing files; one chapter can explain dozens of files without naming each
Repository Encyclopedia (this chapter’s target)	Every significant file: purpose, relationships, entry points, relevance	File- or micro-module entries, prioritized by group A->D below

The book was written around the **architecture**, not around every source file:

Feature Pipeline ↓ CRT ↓ Fusion ↓ Decision ↓ Execution ↓ Risk Gate

...instead of listing file A, file B, file C, That is why only on the order of ~31 src/ files are exactly cited (~7% of the 456-file src inventory), while loose package orientation reaches roughly 60% of src/ - the major concepts and production spine, not every helper.

If the long-term goal is “100% repository understanding”: treat the current book as roughly 60-70% of the knowledge (concept + spine depth), **not** because 60-70% of files were named, but because the major architectural concepts and production path were captured. The remaining work is the **implementation surfaces** under those concepts.

1. Not covered by the book (~403 files)

From the coverage report (exact NOT_IN_BOOK rows against the functionality Excels):

Area	Not in Book
src/	189
scripts/	214
Total	403

This does **not** mean they are useless. It only means the book never discussed them as named files. Many still sit under packages the book *oriented* (DIR_ORIENTED) - research (142 files), features, agent, engines, runtime - where the chapter explains the system without enumerating every module.

Rule: NOT_IN_BOOK != dead code. Code and active config win on “is this live?”

2. They naturally split into four groups

A. Core but omitted (highest priority)

These are the most important omissions: the book explains the architecture but not every **implementation** file in the spine packages.

Package	Inventory files	Exact cited	Not in book (exact)	Notes
config_layer/	27	7	20	CRT helpers, routers, scorers, builders beyond the named few
core/	20	4	16	Fusion/decision/risk are cited; many orchestration helpers not
control_plane/	11	2	9	Beyond registry.py / server.py
execution/	5	1	4	Beyond loop.py (alerts, overrides, ...)
engines/	13	6	0*	Most engines <i>named</i> ; support modules often DIR-only

runtime/	10	2	0*	backtest_v2 / live hook oriented; internals not file-listed
features/	28	2	0*	Pipeline + schema cited; registry/math surface oriented

* “0 not-in-book” can still mean **DIR_ORIENTED** (package mentioned, individual files not deep-read).

Encyclopedia priority: future chapters or encyclopedia entries for remaining core implementation files, engine support, planner helpers, runtime internals.

Should become: future book chapters *or* encyclopedia micro-entries under Part IV-V packages.

B. Sidecar / support modules

These support the system but are not required to understand the trading spine on day one.

Examples from the coverage report (mostly 0 exact cites, full package unread at file level):

Package	Files	Role
bitnet/	23	Low-bit model surface (inert on active config when disabled)
utils/	17	Shared helpers
retrieval/	9	Retrieval helpers
training/	8	Training pipelines
expansion/	6	Parameter expansion / search
portfolio/	6	Portfolio allocation (built; spine still largely single-candle)
replay/	6	Replay harness support
analytics/	5	Analytics
regime/	4	Regime observers (research-adjacent)
events/	2	Event utilities
search/	1	Search helpers
msip/	6	Shadow market-state (oriented in Ch.18; not spine)
multi_llm/	5	Multi-LLM coordination runtime (Ch.23 orients protocol)

Encyclopedia priority: medium - document when a task touches them; do not block spine literacy.

C. Dormant / archived

The book intentionally deprioritized these because they are not part of the active production story (or live only as archive).

Package	Files	Why deprioritized
strategies/	18	S01-S10 wrappers; mostly scaffold / non-spine
scanner/	6	Sidecar opportunity scanning
journal/	6	Journal path; dormant class
data_ingestion/	6	Alternate ingest surface
llm_research/	5	LLM research experiments
uat/	4	UAT
cognitive/	2	Cognitive bus - sidecar (F-012 class)
feedback/	2	Feedback loop - dormant class
monitoring/	2	Monitoring sidecar
ui/	1	UI remnants
archive/inout_legacy/	(archived tree)	Parallel execution rail retired 2026-05-02 (Ch.15)

Encyclopedia priority: low - one-line “status: DORMANT / ARCHIVED + replacement path” is enough unless reopening.

D. Operational scripts (largest numeric gap)

214 scripts are NOT_IN_BOOK (of 335 in `scripts_business_functionality.xlsx`). Only ~4 scripts are exactly cited. The book focuses on **what the system is**, not how every utility script works.

Typical script categories (from the scripts functionality inventory + book intent):

- migrations and repair tools
- validation utilities
- report generators
- governance / census / certification runners
- one-off research runners (`scripts/research/**`)
- data preparation and fetch wrappers
- automation / maintenance

Encyclopedia priority: index by **category + entry point + when to run**, not a narrative chapter per script. Prefer linking `docs/reference/cli-matrix.md` and `docs/reference/script-matrix.md` (when present) as the machine index; encyclopedia entries add purpose and danger notes (write vs read-only).

3. Why only ~35 files named?

Because the book teaches **flow and purpose**:

Concept chapter	Explains without listing every file under...
Ch.07 Feature pipeline	src/features/ (28 files)
Ch.08 CRT	most of crt_engine_v2 surface
Ch.10 Engines	src/engines/ (13 files)
Ch.11-14 Decision / execution / risk	core spine modules
Ch.19-20 Research	src/research/ (142 files) as a map, not 142 essays

One chapter can explain the purpose of **dozens** of files without naming each one individually. That is a feature of an architecture guide - and a **gap** for an encyclopedia.

4. Target: true Repository Encyclopedia

Transform the knowledge book from an **architecture guide** into a **repository encyclopedia**, where essentially every significant file has a documented:

Field	Meaning
Purpose	Why the file exists (user intent / economic role if any)
Relationships	Imports / callers / callees (point at module-roles / code-map when generated)
Entry points	CLI, tests, agent tools, control-plane commands
Relevance	Group A/B/C/D + LIVE / SIDECAR / DORMANT / ARCHIVED
Authority	Research-only vs production-touching; config keys if any
Coverage	CITED / DIR_ORIENTED / NOT_IN_BOOK from the coverage report

Out of encyclopedia scope (still real files): pure __pycache__, generated binaries, bulk market CSV/XLSX data dumps, and duplicate archives - index existence only, do not narrate.

Recommended build order (priority)

Phase	Scope	Outcome	Status
E0	This charter + coverage Excel	Shared vocabulary A-D	DONE
E1	Group A spine packages (core, config_layer, runtime, execution, engines, live, inout, control_plane)	File-level purpose map for ~94 modules	DONE - encyclopedia/E1-spine-implementation
E1b	Deep src/features/ (pipeline + ontology registry, 28 modules)	Feature math authority map	DONE - encyclopedia/E1b-features-registry

E2	Research utilities under src/research/ (143) + scripts/research/ (126) - index, not per-file essays	Research encyclopedia layer	DONE - encyclopedia/E2-research-utilities.
E3	Governance tooling (src/governance/ 19 + scripts/governance/ 31 + scripts/analysis/ 115 + maintenance 9)	Ops + truth maintenance tools	DONE - encyclopedia/E3-governance-tooling.
E4	Group B sidecars (~100 files: bitnet, utils, training, portfolio, replay, analytics, expansion, regime, msip, multi_llm, retrieval, ...)	Support library map	DONE - encyclopedia/E4-sidecar-modules.md
E5	Group C dormant / archived (~69 src/ + archive/**)	Honest “do not reopen without intent”	DONE - encyclopedia/E5-dormant-modules.md
E6	Remaining operator scripts (~70: data/training/context/backtest/export/...)	Operator encyclopedia	DONE - encyclopedia/E6-remaining-scripts.m

E1 note: Architecture chapters still own deep narrative for CITED spine modules (EngineRunner, CRT, Fusion, ...).
E1 fills the **omitted implementation files** so Group A is no longer a blank space on the coverage map.

Entry format (proposed template for future encyclopedia rows)

```
### `src/path/to/module.py` - **Group:** A | B | C | D - **Relevance:** LIVE | SIDECAR | DORMANT |  
ARCHIVED - **Purpose:** ... - **Entry points:** ... - **Related chapters:** ... - **Config keys:** ... |  
none - **Tests:** ... - **Do not confuse with:** ...
```

Machine twin (future, optional): append-only JSONL under docs/book/encyclopedia/ or a generated
sheet sibling to Book_PDF_File_Coverage_Grok.xlsx - **generated from inventory + hand overlays**,
never hand-edited mass dumps as sole truth.

5. Relationship to existing record systems

This encyclopedia does **not** replace:

System	Still owns
Code	Behavior
docs/topics/	Always-synced concept dossiers
docs/memory/*	Edit-time reading order
docs/current-findings.md	Validated conclusions
Functionality Excels	File inventory + summaries
Coverage Excel	Book citation status
CLI / script matrices	How to run X

The encyclopedia **joins** them: file -> purpose -> chapter -> tests -> run entry.

Classification

Concept	Status
Canonical Knowledge Book (Ch.00-23)	Production architecture guide
Remaining-file groups A-D	Charter - authoritative prioritization for future work
Full prioritized encyclopedia	E0-E6 + E1b + JSONL DONE
E1 spine implementation map	encyclopedia/E1-spine-implementation.md
E1b features / registry map	encyclopedia/E1b-features-registry.md (28 modules)
E2 research utilities index	encyclopedia/E2-research-utilities.md (269 files)
E3 governance tooling index	encyclopedia/E3-governance-tooling.md (~174 files)
E4 sidecar modules map	encyclopedia/E4-sidecar-modules.md (~100 files)
E5 dormant / archived map	encyclopedia/E5-dormant-modules.md (~69 + archive)
E6 remaining operator scripts	encyclopedia/E6-remaining-scripts.md (~70 scripts)
Machine JSONL twin	encyclopedia/encyclopedia_rows.jsonl (~813 rows; regenerate via <code>grok/build_encyclopedia_jsonl.py</code>)
Coverage report	Generated tooling (<code>grok/book_file_coverage_report.py</code>)

Authoritative sources

- `grok/Book_PDF_File_Coverage_Grok.xlsx` - CITED / DIR / NOT_IN_BOOK counts.
- `results/analysis/src_business_functionality.xlsx` - 456 src files.
- `scripts_business_functionality.xlsx` - 335 scripts.
- `docs/book/README.md` - book TOC.
- `docs/architecture/module-roles.generated.md` - one-line roles when regenerated.
- `docs/reference/cli-matrix.md` / script matrix - runnable surfaces.
- [Chapter 04](#), [Chapter 15](#), [A2](#).

Unresolved questions

- Whether encyclopedia entries live only as chapters, as a JSONL registry, or both (recommended: hand-written priority entries + generated inventory join).
- Whether Group D should be 100% prose or mostly matrix + danger tags (recommended: matrix-first).
- E1 Group A map is written; optional **E1b** for full `src/features/` file rows remains open.
- **E0-E6 + E1b + JSONL twin complete**. Optional later: richer per-row relationship graphs; agent query UI over JSONL.

Previous: [Chapter 23 - Multi-LLM Coordination](#) * **Next:** [Appendix A1 - Testing](#) (or continue encyclopedia E1)

Related: [Chapter 00 - Quick Start](#) * [Chapter 04 - Architecture at a Glance](#) * [A2 - Unresolved Questions](#)

Memory: `docs/memory/architecture-memory.md` before expanding spine package entries.

Repository Encyclopedia

Parent charter: [Chapter 24 - Repository Encyclopedia](#)

Coverage report: [grok/Book_PDF_File_Coverage_Grok.xlsx](#)

Purpose: File-level purpose, relationships, entry points, and relevance - complementing the architecture book (Ch.00-23), not replacing code.

Status by phase

Phase	Scope	Status	Document
E0	Charter + Groups A-D vocabulary	DONE	Ch.24
E1	Group A spine implementation (core, config_layer, runtime, execution, engines, live, inout, control_plane)	DONE (file map)	E1-spine-implementation.md
E1b	Deep src/features/ pipeline + registry (28 modules)	DONE	E1b-features-registry.md
E2	Research utilities (src/research/ 143 + scripts/research/ 126)	DONE (index)	E2-research-utilities.md
E3	Governance tooling (src/governance + gov/analysis/maintenance scripts)	DONE (index)	E3-governance-tooling.md
E4	Group B sidecars (~100 files: bitnet, utils, training, portfolio, ...)	DONE	E4-sidecar-modules.md
E5	Group C dormant / archived (~69 files + archive/)	DONE	E5-dormant-modules.md
E6	Remaining operator scripts (~70 outside research/gov/analysis/maintenance)	DONE	E6-remaining-scripts.md
JSONL	Machine twin of all encyclopedia module rows	GENERATED	encyclopedia_rows.jsonl

Series status: E0-E6 + E1b + JSONL twin complete.

Machine twin (JSONL)

Item	Value
Artifact	encyclopedia_rows.jsonl (~813 rows)
Regenerate	<code>python grok/build_encyclopedia_jsonl.py</code>
Validate	<code>python grok/build_encyclopedia_jsonl.py --check</code>

Schema (per line)	id, path, kind, phase, group, relevance, purpose, package, classes, source_doc, book_status, bytes, generated_at
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Do not hand-edit the JSONL as sole truth - regenerate from the builder (mirrors GENERATED tier discipline).
Enrichment lives in the markdown encyclopedias + builder overrides.

How to read an entry

Each significant file has:

Field	Meaning
Group	A (core) * B (sidecar) * C (dormant) * D (scripts)
Relevance	LIVE * SIDECAR * DORMANT * ARCHIVED * OBSERVE_ONLY * PARTIAL
Book	CITED * DIR_ORIENTED * NOT_IN_BOOK (from coverage report)
Purpose	Why it exists
Relationships	Who calls it / what it calls
Entry points	CLI / tests / agent / control plane
Chapter	Architecture chapter if any

Code always wins on conflict with this encyclopedia.

Inventories used

- results/analysis/src_business_functionality.xlsx
- scripts_business_functionality.xlsx
- grok/Book_PDF_File_Coverage_Grok.xlsx (regenerate: python grok/book_file_coverage_report.py)

Encyclopedia E1 - Spine Implementation (Group A)

Phase: E1

Status: DONE (file-level map; not line-by-line reverse engineering)

Date: 2026-08-07

Parent: [Chapter 24](#) * [Encyclopedia index](#)

Scope

Group A packages that implement or host the candle -> order spine and its immediate I/O / control surfaces:

Package	Files map	Architecture chapters
src/core/	20	Ch.11-14, Ch.02
src/config_layer/	28 (incl. rr/)	Ch.06-08, Ch.13, Ch.16
src/engines/	13	Ch.10
src/runtime/	10	Ch.05, Ch.15, Ch.02
src/execution/	5	Ch.15
src/live/	3	Ch.15
src/inout/	4	Ch.05, Ch.15
src/control_plane/	11	Ch.22
Total	~94	

Not in E1: full src/features/ registry depth (package oriented in Ch.06-07; file encyclopedia deferred to a follow-on spine-features pass if needed), src/research/ (E2), governance (E3).

Legend - Relevance

Tag	Meaning
LIVE	On or adjacent to production spine / promotion path
PARTIAL	Built + tested; wiring conditional or incomplete
SIDECAR	Optional / advisory / non-veto on default path
OBSERVE_ONLY	Telemetry or probes; must not change decisions
SHIM	Compatibility alias; prefer canonical module
INERT	Present but disabled / unwired on active config

1. `src/core/` - Decision spine orchestration

Package role: Orchestrate scoring -> fusion -> decision -> capital gate; hold shared types and optional controllers.

Already in the architecture book (CITED)

File	Classes	Purpose (short)	Chapter
<code>engine_runner.py</code>	<code>EngineRunner</code>	Orchestrates engines, fusion, decision path	Ch.02, Ch.10-12
<code>fusion_engine.py</code>	<code>FusionEngine</code> , ...	Multi-engine score fusion / completeness	Ch.11
<code>decision_engine.py</code>	<code>DecisionEngine</code>	Semantic accept/reject (no economic RR)	Ch.12
<code>ultron_risk_gate.py</code>	<code>UltronRiskGate</code>	Final capital / RR / exposure check	Ch.14

Encyclopedia entries - omitted / supporting

`src/core/__init__.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Package exports for the decision spine.
- **Relationships:** Imported by runtime, agents, tests.
- **Entry points:** Import surface only.
- **Chapter:** Ch.04 package map.

`src/core/types.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Canonical TypedDict / contract types for the execution spine (`EngineContext`, engine results, etc.).
- **Relationships:** Consumed by `EngineRunner`, engines, tests that pin shapes.
- **Entry points:** Imported across `src/core`, `src/engines`, tests.
- **Chapter:** Complements `docs/reference/schemas.md`.

`src/core/feature_store.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Canonical feature store - single source of truth for feature dictionaries used on the spine.
- **Relationships:** Fed by feature builders / pipeline; read by engines and gates.
- **Entry points:** `EngineRunner` / live hook paths; unit tests.
- **Chapter:** Ch.07 (vector) without naming this module.

`src/core/gate_intelligence.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Pure signal gate used by `ExecutionPlanner` (intent / geometry-adjacent approval), not the capital gate.
- **Relationships:** Called from `execution_planner.py`; uses CRT-level geometry helpers (`compute_crt_levels` chain).
- **Entry points:** Planner path after Decision ACCEPT; tests under planner/gate.
- **Chapter:** Ch.13 (planner) - this file is the missing named helper.

- **Note:** ATR unit issues in CRT levels surface here via geometry (Ch.13 open defect).

`src/core/model_registry.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Versioned model registry with promotion margin / GOV-style quality gates for model artifacts.
- **Relationships:** Training / promotion paths; loads model JSON under `models/`.
- **Entry points:** Training scripts, promotion flows; `tests` around registry.
- **Chapter:** Ch.16 (promotion) adjacent.

`src/core/collector.py`

- **Group:** A * **Relevance:** SIDECAR * **Book:** NOT_IN_BOOK
- **Purpose:** Append structured records to `logs/collector.jsonl` (telemetry for fusion shadow / analysis).
- **Relationships:** Written from EngineRunner / runtime; read by `runtime/analyze_fusion_shadow.py`.
- **Entry points:** Runtime with collector enabled; analysis scripts.
- **Chapter:** None (ops telemetry).

`src/core/acceptance_controller.py`

- **Group:** A * **Relevance:** SIDECAR / PARTIAL * **Book:** NOT_IN_BOOK
- **Purpose:** Adaptive acceptance control - modulates accept rates / thresholds over time.
- **Relationships:** Optional layer near fusion/decision; config-driven when wired.
- **Entry points:** Config sections if present; focused tests.
- **Chapter:** None - not required to understand baseline spine.

`src/core/convergence_controller.py`

- **Group:** A * **Relevance:** SIDECAR / PARTIAL * **Book:** NOT_IN_BOOK
- **Purpose:** Convergence layer for FusionEngine (stability of multi-engine agreement).
- **Relationships:** Fusion path when enabled.
- **Entry points:** Fusion config flags; tests.
- **Chapter:** Ch.11 optional depth.

`src/core/dynamic_threshold.py`

- **Group:** A * **Relevance:** SIDECAR * **Book:** NOT_IN_BOOK
- **Purpose:** Percentile-based adaptive thresholds (not fixed hardcodes).
- **Relationships:** Controllers / gates that need regime-aware cutoffs.
- **Entry points:** Config + unit tests.
- **Chapter:** None.

`src/core/hierarchical_meta_fusion.py`

- **Group:** A * **Relevance:** SIDECAR / INERT-ish * **Book:** NOT_IN_BOOK
- **Purpose:** Multi-layer hierarchical meta-fusion for capital allocation quality scoring (HMF).
- **Relationships:** Sidecar to main fusion; findings class F-012 (sidecar-only consumers).
- **Entry points:** Explicit HMF callers / research; not default spine.
- **Chapter:** Ch.04 dormant/sidecar class.

`src/core/regime_governor.py`

- **Group:** A * **Relevance:** SIDECAR * **Book:** NOT_IN_BOOK

- **Purpose:** Regime-aware governor (`UltronGovernor` naming in module) - regime context for sizing/gates.
- **Relationships:** Adjacent to Ultron / risk; not a substitute for `UltronRiskGate`.
- **Entry points:** Config-gated integrations; tests.
- **Chapter:** Ch.14 adjacent.

`src/core/signal_audit.py`

- **Group:** A * **Relevance:** OBSERVE_ONLY * **Book:** NOT_IN_BOOK
- **Purpose:** Per-bar signal audit + leak detection recorder.
- **Relationships:** Observes spine outputs; should not mutate decisions.
- **Entry points:** Audit-enabled runs; analysis.
- **Chapter:** None.

`src/core/signal_belief_tracker.py`

- **Group:** A * **Relevance:** PARTIAL * **Book:** NOT_IN_BOOK
- **Purpose:** Temporal belief accumulator that can gate `DecisionEngine` calls (multi-bar confidence).
- **Relationships:** Upstream of `DecisionEngine` when enabled.
- **Entry points:** Config + tests; may be inactive on default configs.
- **Chapter:** None.

`src/core/governance_mode.py`

- **Group:** A * **Relevance:** LIVE (LLM layer) * **Book:** NOT_IN_BOOK
- **Purpose:** `GOVERNANCE_MODE` switch hardening for the LLM layer (Trd-M5) - when LLM is advisory vs restricted.
- **Relationships:** LLM client / agent governance; not a trade veto by itself.
- **Entry points:** Config / env governance mode; agent paths.
- **Chapter:** Ch.21 / Ch.16 adjacent.

`src/core/backtest_port.py`

- **Group:** A * **Relevance:** LIVE (interface) * **Book:** NOT_IN_BOOK
- **Purpose:** Dependency-inversion port for backtest results (Trd-M4) - clean boundary between runtime and consumers.
- **Relationships:** Implemented/used by backtest harnesses and validators.
- **Entry points:** Backtest / config validator.
- **Chapter:** Ch.05 / Ch.16.

`src/core/ultron_risk_gate_wrapper.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Wrapper around `ultronRiskGate` for alternate call shapes / integration adapters.
- **Relationships:** Prefer `ultron_risk_gate.py` as canonical; wrapper for callers needing a facade.
- **Entry points:** Live / strategy adapters; tests.
- **Chapter:** Ch.14.

2. `src/config_layer/` - Config, CRT, planner, models, LLM transport

Package role: Production config load, CRT state machine, execution planner, goal schema, model path resolution, RR offline stack, optional LLM transport.

Already in the architecture book (CITED)

File	Purpose	Chapter
<code>crt_engine_v2.py</code>	Full CRT state machine	Ch.08
<code>execution_planner.py</code>	Intent classifier + gate	Ch.13
<code>config_validator.py</code>	Pre-promotion validation	Ch.16
<code>goal_schema.py</code> / <code>goal_validator.py</code>	Goal layer	Ch.01
<code>state_identity.py</code>	CRTState / transitions seed	Ch.08
<code>crt_gaussian_scorer.py</code>	CRT-side Gaussian scoring helper	Ch.10 adjacent

Encyclopedia entries

`src/config_layer/production_config.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Load active production config; `get_active_version()` / `get_prod_config()` - **Tier-0 runtime truth** (`ACTIVE_VERSION`).
- **Relationships:** Nearly every production path; fail-fast if version missing.
- **Entry points:** Import from engines, backtest, promotion, CLI.
- **Chapter:** CLAUDE.md §4.0; Ch.16.

`src/config_layer/config_builder.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Single source of truth for building `CRTConfig` (and override validation). Site of F-057 split-brain when programmatic path skips production JSON.
- **Relationships:** `market_router` profiles; `BacktestRunner` / CRT engine.
- **Entry points:** Backtest, tuner, tests; CLI path usually via production JSON.
- **Chapter:** Ch.13 (F-057), Ch.08.

`src/config_layer/market_router.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Market-type CRT profiles (crypto/forex) and routing into `ConfigBuilder` defaults.
- **Relationships:** Feeds `ConfigBuilder` when production overrides absent.
- **Entry points:** `ConfigBuilder`; instrument onboarding.
- **Chapter:** Ch.13 F-057 mechanism.

`src/config_layer/production_bundle.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Immutable executable production state as one object (bundle of config + identity members).
- **Relationships:** Promotion / runtime loaders that want a sealed snapshot.
- **Entry points:** Governance / production load paths.
- **Chapter:** Ch.16.

`src/config_layer/stack_version.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Composite identity over WHAT / HOW / WHO / EXECUTION stack versions.

- **Relationships:** Bundle / promotion identity checks.
- **Entry points:** Governance tooling.
- **Chapter:** Ch.16 adjacent.

`src/config_layer/model_paths.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** CODE layout authority for where model artifacts live on disk.
- **Relationships:** `model_resolver`, registries under `models/`.
- **Entry points:** Trainers, resolvers, EngineRunner model load.
- **Chapter:** Ch.10 / Ch.16.

`src/config_layer/model_resolver.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Resolve model family -> registry active entry -> artifact path (+ identity parity).
- **Relationships:** Zone / RR / Gaussian / BitNet loaders.
- **Entry points:** Engine construction; tests for registry parity (e.g. zone manifest).
- **Chapter:** Ch.10, F-041 class.

`src/config_layer/state_topology.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Immutable CRT state topology view (`VALID_TRANSITIONS` runtime topology).
- **Relationships:** Breaks import cycles vs full `crt_engine_v2`; used by tests and resolvers.
- **Entry points:** CRT tests (`test_crt_state_invariants`), topology consumers.
- **Chapter:** Ch.08.

`src/config_layer/state_contract.py` / `state_contract_loader.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Typed immutable CRT state contract + loader for declarative/YAML-backed contract surfaces.
- **Relationships:** Semantic/state resolver pipelines; CRT governance.
- **Entry points:** Contract tests; semantic pipeline.
- **Chapter:** Ch.08 / Ch.06 adjacent.

`src/config_layer/crt_config_provenance.py`

- **Group:** A * **Relevance:** OBSERVE_ONLY / LIVE-meta * **Book:** NOT_IN_BOOK
- **Purpose:** Track how a CRTConfig was constructed (construction mode) - observability for F-057 class bugs.
- **Relationships:** ConfigBuilder / BacktestRunner diagnostics.
- **Entry points:** Probes and governance checks.
- **Chapter:** Ch.13 F-057.

`src/config_layer/crt_sweep_taxonomy.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Pure-function geometric classification of CRT sweeps (taxonomy helpers for engine/research).
- **Relationships:** CRT engine and research classifiers.
- **Entry points:** CRT tests; research scripts.
- **Chapter:** Ch.08 adjacent.

`src/config_layer/llm_inference_client.py`

- **Group:** A * **Relevance:** SIDECAR (fail-open) * **Book:** NOT_IN_BOOK
- **Purpose:** LLM HTTP transport with timeout / failure disable; returns neutral scores after N failures.
- **Relationships:** Agent, optional gates; **not** hard dependency of trade path.
- **Entry points:** Agent, llama/LLM config sections.
- **Chapter:** CLAUDE.md constraints; Ch.21.

`src/config_layer/llm_scorer.py / llm_narrative.py / insight_reporter.py`

- **Group:** A * **Relevance:** SIDECAR * **Book:** NOT_IN_BOOK
- **Purpose:** Score computation, narrative generation, and insight reporting over LLM transport.
- **Relationships:** Optional engines (`llm_engine`), dashboards, reports.
- **Entry points:** Control-plane report APIs; optional engine path.
- **Chapter:** Ch.21 / Ch.22 adjacent.

`src/config_layer/rr/rr_dataset_builder.py`

- **Group:** A * **Relevance:** RESEARCH / LIVE-train * **Book:** NOT_IN_BOOK
- **Purpose:** Build RR training datasets from features + outcomes (label path historically F-022-sensitive).
- **Relationships:** `rr_pattern_miner`, training scripts.
- **Entry points:** `scripts/training/*`, research.
- **Chapter:** Ch.10 RR lineage; F-022/F-045.

`src/config_layer/rr/rr_pattern_miner.py`

- **Group:** A * **Relevance:** RESEARCH / ARTIFACT * **Book:** NOT_IN_BOOK
- **Purpose:** Train / pure-Python inference for RR pattern model (Mahalanobis-style confidence).
- **Relationships:** Produces `models/rr_model*.json`; F-044 gate mis-spec documented in findings.
- **Entry points:** Training scripts; offline probes.
- **Chapter:** Ch.10; F-044/F-038.

`src/config_layer/rr/rr_fusion.py`

- **Group:** A * **Relevance:** INERT on active (disabled) * **Book:** NOT_IN_BOOK
- **Purpose:** Advisory RR fusion layer; **disabled** in active production config (`rr_fusion.enabled: false`) so base RREngine flows unmutated.
- **Relationships:** EngineRunner only constructs when enabled.
- **Entry points:** Config flag; regression tests for disabled identity.
- **Chapter:** Ch.10-11; F-038/F-044.

`src/config_layer/rr/__init__.py`

- **Group:** A * **Relevance:** LIVE (package) * **Book:** NOT_IN_BOOK
- **Purpose:** RR subpackage marker / exports.
- **Chapter:** Ch.10.

`src/config_layer/__init__.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Package docstring / exports for config layer.
- **Chapter:** Ch.04.

3. `src/engines/` - Scoring engines + gates

Package role: Four mandatory engines (CRT, Gaussian, Zone Gate, RR) plus optional validators and meta layers.

Already CITED

`crt_engine.py`, `heuristic_gaussian_engine.py`, `ml_gaussian_engine.py`, `rr_engine.py`, `scoring_engine.py`, `zone_gate_engine.py` - see **Ch.10**.

Encyclopedia entries - support / optional

`src/engines/__init__.py`

- **Group:** A * **Relevance:** LIVE * **Book:** DIR_ORIENTED
- **Purpose:** Package exports for scoring engines.
- **Chapter:** Ch.10.

`src/engines/gaussian_engine.py`

- **Group:** A * **Relevance:** SHIM * **Book:** DIR_ORIENTED
- **Purpose:** Backward-compatibility shim; canonical implementations are heuristic / ML Gaussian modules.
- **Relationships:** Prefer `heuristic_gaussian_engine.py` / `ml_gaussian_engine.py`.
- **Chapter:** Ch.10.

`src/engines/live_engine.py`

- **Group:** A * **Relevance:** LIVE * **Book:** DIR_ORIENTED
- **Purpose:** Live trading decision helper + zone load path used by live hook (not the full HookedLiveEngine).
- **Relationships:** `live_engine_hook` subclasses / wraps live engine behaviors; zone registry load.
- **Entry points:** Live path; tests.
- **Chapter:** Ch.15.

`src/engines/zone_cluster_score.py`

- **Group:** A * **Relevance:** LIVE * **Book:** DIR_ORIENTED
- **Purpose:** Pure zone cluster scoring shared by EngineRunner and historical zone tooling.
- **Relationships:** Zone gate / `discover_zones` pipelines.
- **Chapter:** Ch.10 Zone Gate.

`src/engines/trap_validator_engine.py`

- **Group:** A * **Relevance:** PARTIAL * **Book:** DIR_ORIENTED
- **Purpose:** Data integrity / trap validation pre-gate (quality of setup, not fusion weight).
- **Relationships:** Optional pre-filter before or beside scoring.
- **Entry points:** Config when enabled; tests.
- **Chapter:** None dedicated.

`src/engines/llm_engine.py`

- **Group:** A * **Relevance:** SIDECAR * **Book:** DIR_ORIENTED
- **Purpose:** Parallel engine wrapper around `llm_score_safe()` - advisory, fail-open class.
- **Relationships:** `llm_scorer` / inference client.
- **Chapter:** Ch.21 constraints (LLM not hot-path hard dep).

`src/engines/tradenet_meta_engine.py`

- **Group:** A * **Relevance:** INERT / UNWIRED * **Book:** DIR_ORIENTED
- **Purpose:** TradeNet meta-cognition layer - fusion neural slot historically stubbed (F-005).
- **Relationships:** Not on active production path without TN qualification protocol.
- **Chapter:** Ch.10 / F-005; tradenet lineage audit.

4. src/runtime/ - Backtest, live hook, baselines, probes

Package role: Run the spine over historical or live candles; capture baselines; optional observation tools.

Already CITED

File	Purpose	Chapter
backtest_v2.py	Candle-by-candle backtest harness	Ch.05, Ch.02
live_engine_hook.py	HookedLiveEngine.process live tick entry	Ch.15

Encyclopedia entries

src/runtime/__init__.py

- **Group:** A * **Relevance:** LIVE * **Book:** DIR_ORIENTED
- **Purpose:** Package marker.
- **Chapter:** Ch.04.

src/runtime/baseline_capture.py

- **Group:** A * **Relevance:** LIVE (governance hygiene) * **Book:** DIR_ORIENTED
- **Purpose:** Capture reproducible baseline manifest (schema hash, models, config SHA).
- **Relationships:** Pre-training / pre-migration hygiene.
- **Entry points:** python src/runtime/baseline_capture.py --label ...
- **Chapter:** CLAUDE.md Phase 0; Ch.16.

src/runtime/unified_replay_harness.py

- **Group:** A * **Relevance:** LIVE (research/ops) * **Book:** DIR_ORIENTED
- **Purpose:** Run execution-truth backtest (v2) and BitNet gate modes on same dataset; emit comparable artifacts.
- **Relationships:** Backtest + BitNet shadow configs.
- **Entry points:** Research / replay CLIs; determinism tests.
- **Chapter:** Ch.05; F-055 class.

src/runtime/backtest_bitnet.py

- **Group:** A * **Relevance:** RESEARCH / SHADOW * **Book:** DIR_ORIENTED
- **Purpose:** Row-by-row backtest with BitNet as hard gatekeeper (shadow experiments).
- **Relationships:** BitNet package; shadow production configs.
- **Entry points:** Research scripts; F-055 shadow.
- **Chapter:** Ch.10 BitNet; F-004/F-055.

src/runtime/analyze_fusion_shadow.py

- **Group:** A * **Relevance:** OBSERVE_ONLY * **Book:** DIR_ORIENTED

- **Purpose:** Analyze EngineRunner fusion shadow telemetry from collector logs.
- **Relationships:** collector.py JSONL.
- **Entry points:** Offline analysis CLI.
- **Chapter:** None.

src/runtime/crt_baseline_trace.py

- **Group:** A * **Relevance:** OBSERVE_ONLY * **Book:** DIR_ORIENTED
- **Purpose:** Behavior-preserving CRT baseline observation / operand traces.
- **Relationships:** CRT engine diagnostics.
- **Entry points:** Forensic / baseline programs.
- **Chapter:** Ch.08 diagnostics.

src/runtime/crt_fail_reason_counters.py

- **Group:** A * **Relevance:** OBSERVE_ONLY * **Book:** DIR_ORIENTED
- **Purpose:** Count CRT try_* fail reasons - observation only.
- **Relationships:** CRT funnel diagnostics.
- **Entry points:** Governance / CRT census tools.
- **Chapter:** Ch.08 / governance CRT artifacts.

src/runtime/exit_model_band.py

- **Group:** A * **Relevance:** RESEARCH * **Book:** DIR_ORIENTED
- **Purpose:** Dual-bound exit-model reporting (trust-layer exit bands).
- **Relationships:** Research exit/cost studies (F-025 class).
- **Entry points:** Research scripts.
- **Chapter:** Ch.19 exit falsification.

5. src/execution/ - Multi-signal execution loop (partial)

Already CITED

loop.py - multi-signal execution loop; **built, tested, no live caller** (Ch.15).

Encyclopedia entries

src/execution/__init__.py

- **Group:** A * **Relevance:** PARTIAL * **Book:** NOT_IN_BOOK
- **Purpose:** Package marker.
- **Chapter:** Ch.15.

src/execution/alert_manager.py

- **Group:** A * **Relevance:** PARTIAL * **Book:** NOT_IN_BOOK
- **Purpose:** Alerts for execution loop events (human-visible notifications).
- **Relationships:** ExecutionLoop; may bridge to Telegram later.
- **Entry points:** Loop when wired; tests.
- **Chapter:** Ch.15.

src/execution/override_handler.py

- **Group:** A * **Relevance:** PARTIAL * **Book:** NOT_IN_BOOK
- **Purpose:** Human/system overrides against automated execution decisions.
- **Relationships:** Execution loop; safety rail.
- **Entry points:** Loop when wired; tests.
- **Chapter:** Ch.15.

`src/execution/execution_intent_v1_0.py`

- **Group:** A * **Relevance:** PARTIAL * **Book:** NOT_IN_BOOK
- **Purpose:** Human-in-the-loop execution intent state machine (v1.0 contract).
- **Relationships:** Distinct from CRT `ExecutionPlanner` intents; for loop-level intent lifecycle.
- **Entry points:** Execution loop / UAT; tests (`test_execution_*`).
- **Chapter:** Ch.15.

6. `src/live/` - Live broker / alert bridges

`src/live/__init__.py`

- **Group:** A * **Relevance:** LIVE * **Book:** DIR_ORIENTED
- **Purpose:** Package marker for live bridges.
- **Chapter:** Ch.15.

`src/live/mt5_bridge.py`

- **Group:** A * **Relevance:** LIVE * **Book:** DIR_ORIENTED
- **Purpose:** MetaTrader 5 order execution bridge (IPC to terminal).
- **Relationships:** Live path after Ultron approve; requires MT5 running.
- **Entry points:** Live engine / operator setup.
- **Chapter:** Ch.15; external MT5.

`src/live/telegram_bridge.py`

- **Group:** A * **Relevance:** SIDECAR * **Book:** DIR_ORIENTED
- **Purpose:** Telegram bot alerts for signals / status.
- **Relationships:** Live engine optional notifications (not capital authority).
- **Entry points:** Live config with bot token (secrets in `.env` - never commit).
- **Chapter:** Ch.15.

7. `src/inout/` - Market data I/O (not the archived strategy rail)

Resolved in Ch.15: fetchers only; archived parallel spine is `archive/inout_legacy/`.

`src/inout/mt5_candle_fetcher.py`

- **Group:** A * **Relevance:** LIVE * **Book:** DIR_ORIENTED
- **Purpose:** Fetch M15 (etc.) OHLCV from MT5 -> CSV for `CandleLoader`.
- **Entry points:** `scripts/data/fetch_candles_mt5.py`.
- **Chapter:** Ch.05, Ch.15.

`src/inout/alphavantage_candle_fetcher.py`

- **Group:** A * **Relevance:** LIVE (data) * **Book:** DIR_ORIENTED
- **Purpose:** Alpha Vantage FX intraday -> CSV (premium limits apply).
- **Chapter:** Ch.05.

`src/inout/hummingbot_candle_fetcher.py`

- **Group:** A * **Relevance:** LIVE (data) * **Book:** DIR_ORIENTED
- **Purpose:** Exchange OHLCV via hummingbot-supported venues -> CSV.
- **Chapter:** Ch.05.

`src/inout/perp_funding_fetcher.py`

- **Group:** A * **Relevance:** RESEARCH / LIVE-data * **Book:** DIR_ORIENTED
- **Purpose:** Binance USD-M perpetual funding/basis inputs for carry research (Programs 6/6b).
- **Chapter:** Ch.19 P6; Ch.05.

8. `src/control_plane/` - Localhost command catalog + dashboard

Already CITED

`registry.py` (CommandSpec catalog), `server.py` (stdlib HTTP localhost:8787) - **Ch.22.**

Encyclopedia entries

`src/control_plane/__init__.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Package exports for control plane.
- **Chapter:** Ch.22.

`src/control_plane/cp_types.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** ArgSpec, CommandSpec, RunRecord dataclasses for the catalog and job history.
- **Relationships:** `registry.py`, `jobs.py`, `server.py`.
- **Chapter:** Ch.22; schemas.

`src/control_plane/jobs.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** JobManager - queue and run control-plane commands as subprocesses.
- **Relationships:** Server API -> subprocess argv from CommandSpec.
- **Entry points:** Dashboard “run command”; localhost only.
- **Chapter:** Ch.22.

`src/control_plane/dashboard_api.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Read-only data API for the CRT trading dashboard UI.
- **Relationships:** Server routes; log/artifact readers.
- **Chapter:** Ch.22.

`src/control_plane/monitors.py`

- **Group:** A * **Relevance:** LIVE * **Book:** NOT_IN_BOOK
- **Purpose:** Dashboard monitor specs (jsonl-tail / json / regex / file-stat sources).
- **Relationships:** Dashboard live panels.
- **Chapter:** Ch.22.

`src/control_plane/report_api.py`

- **Group:** A * **Relevance:** SIDECAR * **Book:** NOT_IN_BOOK
- **Purpose:** Run history reports (Excel) + optional LLM analysis.
- **Relationships:** Job history; LLM client.
- **Chapter:** Ch.22 adjacent.

`src/control_plane/context_report.py`

- **Group:** A * **Relevance:** SIDECAR * **Book:** NOT_IN_BOOK
- **Purpose:** ContextReportAPI - architecture analyst pack for a run (LLM-assisted).
- **Relationships:** `code_context_extractor`, `dot_graph_context`.
- **Chapter:** Ch.22 / multi-LLM ops.

`src/control_plane/code_context_extractor.py`

- **Group:** A * **Relevance:** SIDECAR * **Book:** NOT_IN_BOOK
- **Purpose:** AST-based code extraction for context reports.
- **Chapter:** Ch.22.

`src/control_plane/dot_graph_context.py`

- **Group:** A * **Relevance:** SIDECAR * **Book:** NOT_IN_BOOK
- **Purpose:** Flow resolution + architectural graph context (`.dot`) for reports.
- **Relationships:** `graph.dot` / pyan outputs when present.
- **Chapter:** Ch.04 / Ch.22.

E1 coverage rollup

Package	Exact CITED (pre-E1 book)	E1 encyclopedia entries added	Residual risk
core	4	16 supporting modules mapped	Controllers may be config-off - verify before assuming LIVE
config_layer	7	21 modules mapped	F-057 / rr_fusion disabled still apply
engines	6	7 support modules mapped	TradeNet/LLM inert unless enabled
runtime	2	8 modules mapped	Observation tools must stay OBSERVE_ONLY
execution	1	4 modules mapped	Loop still unwired to live
live	0	3 modules mapped	Secrets never in docs
inout	0	4 fetchers mapped	!= archived strategy rail

control_plane	2	9 modules mapped	localhost-only, no auth
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E1 exit criterion (charter): every Group A spine-package `.py` file has a purpose/relevance row above (or is a trivial empty `__init__` noted as package marker). **Met.**

Not claimed: full call-graph proof for every optional controller; economic authority for any sidecar.

Next

- **E2** - `src/research/` + research scripts index (category + driver map, not 142 essays).
- Optional **E1b** - `src/features/` file encyclopedia (registry package depth) if spine feature math is the next editing focus.

Related chapters: 00 (`../00-quick-start.md`) * 08 (`../08-crt-state-machine.md`) * 10 (`../10-four-scoring-engines.md`) * 11-15 (`../11-fusion.md`) * 16 (`../16-config-first-and-promotion.md`) * 22 (`../22-control-plane.md`) * [24](#)

Encyclopedia E1b - Feature Pipeline & Registry (deep map)

Phase: E1b
Status: DONE (file-level map of `src/features/` - 28 modules)
Date: 2026-08-07
Parent: [Chapter 24](#) * [E1 spine](#) * [Encyclopedia index](#)
Architecture narrative: Chapter 06 (`../06-market-ontology.md`) * Chapter 07 (`../07-feature-pipeline.md`)

Scope

Surface	Count	Role
<code>src/features/*.py</code> (top-level)	23	Pipeline, schema, semantic layers, builders, monitors
<code>src/features/registry/</code>	5	Authoritative feature-math implementation (no eval)
Total	28	Group A feature surface (was DIR_ORIENTED under E1 only)

Authority model (immutable):

```
configs/formulas/market_ontology.yaml -> DECLARES meaning (WHAT) src/features/registry/* +  
candle_math/derived_math -> IMPLEMENTS math (HOW) src/features/feature_pipeline.py -> EMITS 39-dim  
vector (v4.0) src/features/feature_schema.py -> NAMES / pins CANONICAL_FEATURES
```

Never re-derive formula math in engines/scripts - use `registry / derived_math / pipeline` (F-047, ownership lint).

Book status: `feature_pipeline.py` + `feature_schema.py` **CITED** in Ch.07; all other modules were package-oriented until this E1b map.

Layer map (semantic pipeline)

Layer	Module(s)	Role
L0 OHLC identity	<code>candle_math.py</code> , <code>registry/primitive_registry.py</code>	Immutable geometry (body, wick, range, ratios)
L1 derived scalars	<code>derived_math.py</code> , <code>registry/derived_registry.py</code>	ATR-relative / normalized metrics (FM-0NN)
L1 compositions	<code>registry/composition_registry.py</code>	Named ratios over primitives
Registry facade	<code>formula_registry.py</code> , <code>registry/__init__.py</code> , <code>registry/_loader.py</code>	FORMULA_REGISTRY + ontology load
FM resolve	<code>fm_resolve.py</code>	FM-id -> callable (fail-closed)
Identity	<code>feature_identity.py</code>	<code>feature_id</code> collision / governed identities
Batch pipeline	<code>feature_pipeline.py</code>	Production 39-dim vector builder

Schema pin	<code>feature_schema.py</code> , <code>schema_validator.py</code>	CANONICAL_FEATURES / dim / fail-fast validate
Online builders	<code>feature_builder.py</code> , <code>crt_feature_builder.py</code>	Tick/row -> input_data / CRT->canonical dict
Structure / PIT	<code>causal_structure.py</code>	FC1-A delayed-confirmed swings (no lookahead)
Session / clock	<code>session_classifier.py</code> , <code>broker_clock.py</code>	Session ownership; MT5 UTC correction (F-066)
Semantic L2	<code>feature_states.py</code>	Feature values -> declared states
Semantic L4	<code>market_context.py</code>	States -> market context
Semantic L5	<code>market_shape.py</code>	Context -> recurring shapes
Semantic L5 CRT	<code>crt_state_resolver.py</code>	Feature states -> CRT 9-state resolve (parity with engine)
Semantic L7	<code>model_evidence.py</code>	Model testimony per bar
Contracts	<code>gaussian_schema_contract.py</code>	Gaussian NB schema under v4 migration
Dataset prep	<code>dataset_builder.py</code> , <code>dataset_validator.py</code>	Train/log feature extract + validate
Drift	<code>feature_monitor.py</code>	Rolling distribution / Z-drift on trades
Package	<code>__init__.py</code>	Exports

1. Math authority (immutable primitives + registry)

`src/features/candle_math.py`

- **Group:** A * **Relevance:** LIVE * **Book:** DIR_ORIENTED (E1b deep)
- **Purpose:** Single immutable source for candle geometry identities (body/wick/range/ratios). Mechanism not policy (F-046 class).
- **Relationships:** Used by pipeline, CRT, registry primitives; parity-tested.
- **Chapter:** Ch.06-07; F-046.

`src/features/derived_math.py`

- **Group:** A * **Relevance:** LIVE
- **Purpose:** Scalar implementations of deterministic normalized metrics (disp_strength/retest/... FM forms). Sibling of `candle_math`.
- **Relationships:** `registry/derived_registry.py` maps ontology impl names here (no eval).
- **Chapter:** Ch.06-07; F-047/F-050/F-063.

`src/features/registry/primitive_registry.py`

- **Group:** A * **Relevance:** LIVE
- **Purpose:** Maps ontology primitive impl names -> `candle_math` callables.
- **Chapter:** Ch.06.

`src/features/registry/composition_registry.py`

- **Group:** A * **Relevance:** LIVE
- **Purpose:** Trading-interpretation ratios (named callables over primitives).
- **Chapter:** Ch.06.

`src/features/registry/derived_registry.py`

- **Group:** A * **Relevance:** LIVE
- **Purpose:** Maps derived FM impl names -> `derived_math` callables + signature dispatch.
- **Chapter:** Ch.06.

`src/features/registry/_loader.py`

- **Group:** A * **Relevance:** LIVE
- **Purpose:** Leaf ontology YAML loader (lazy PyYAML); avoids import cycles.
- **Relationships:** Reads `configs/formulas/market_ontology.yaml`.
- **Chapter:** Ch.06.

`src/features/registry/__init__.py`

- **Group:** A * **Relevance:** LIVE
- **Purpose:** Authoritative aggregated feature-math registry API (`FORMULA_REGISTRY`, validation helpers).
- **Chapter:** Ch.06-07; F-047.

`src/features/formula_registry.py`

- **Group:** A * **Relevance:** LIVE
- **Purpose:** Back-compat facade re-exporting the registry package (do not re-implement math here).
- **Chapter:** Ch.06.

`src/features/fm_resolve.py`

- **Group:** A * **Relevance:** LIVE
- **Purpose:** Phase-2 FM resolution: FM-id -> callable via ontology + `FORMULA_REGISTRY` (Option A).
- **Entry points:** Certification tooling, consumers needing FM-id dispatch.
- **Chapter:** Ch.06; E3 `feature_dag` tools.

`src/features/feature_identity.py`

- **Group:** A * **Relevance:** LIVE
- **Purpose:** Fail-closed `feature_id` lookup; governs duplicate/collision families (phase-1 identity registry docs).
- **Chapter:** Ch.07 governance adjacent.

2. Production vector emission & schema

`src/features/feature_pipeline.py`

- **Group:** A * **Relevance:** LIVE * **Book:** CITED (Ch.07)
- **Purpose:** Production-grade M15 OHLCV -> deterministic vector of length `CANONICAL_FEATURE_DIM` (39, v4.0). Config-driven indicator periods (`feature_pipeline` section).
- **Relationships:** Backtest + live feature build; BitNet/Zone/CRT consumers of vector slots.
- **Chapter:** Ch.07; F-061/F-066 config gates (`normalization_basis`, `session_timestamp_basis`).

`src/features/feature_schema.py`

- **Group:** A * **Relevance:** LIVE * **Book:** CITED (Ch.07)
- **Purpose:** Canonical feature names, dim, schema objects, hash pins. Source of truth for 39-dim order.
- **Note:** Reference docs may lag (38 vs 39) - code wins.
- **Chapter:** Ch.07; F-062.

`src/features/schema_validator.py`

- **Group:** A * **Relevance:** LIVE
- **Purpose:** Strict fail-fast validator for the canonical feature contract.
- **Entry points:** Dataset/train preflight; tests.
- **Chapter:** Ch.07.

`src/features/gaussian_schema_contract.py`

- **Group:** A * **Relevance:** LIVE-contract
- **Purpose:** Gaussian NB feature-schema contract under SCHEMA-V4-VECTOR-MIGRATION (dim renames).
- **Chapter:** Ch.10 Gaussian; F-062 class.

3. Online / CRT builders

`src/features/feature_builder.py`

- **Group:** A * **Relevance:** LIVE
- **Purpose:** Constructs canonical `input_data` dict from raw tick/row; enforces mandatory OHLCV.
- **Relationships:** Live/engine paths needing lightweight build vs full batch pipeline.
- **Chapter:** Ch.07 adjacent.

`src/features/crt_feature_builder.py`

- **Group:** A * **Relevance:** LIVE / PARTIAL
- **Purpose:** CRT trade/candle/state -> dict with **exactly** all CANONICAL_FEATURES. Historical `body_ratio` outlier path audited (F-046 - prefer `candle_math`).
- **Chapter:** Ch.07-08.

`src/features/causal_structure.py`

- **Group:** A * **Relevance:** LIVE
- **Purpose:** FC1-A delayed-confirmed structure publication helpers (PIT-safe swings). Batch authority via pipeline `structure/liquidity` columns.
- **Chapter:** Ch.07; F-051 class.

4. Session & broker clock

`src/features/session_classifier.py`

- **Group:** A * **Relevance:** LIVE
- **Purpose:** **Single owner** of “what session is this?” - deliberately separates feature session labels vs CRT filter windows (F-066 doctrine).
- **Chapter:** Ch.07; F-017/F-066.

`src/features/broker_clock.py`

- **Group:** A * **Relevance:** LIVE (config-gated)
- **Purpose:** Convert MT5 broker-server timestamps to true UTC when `session_timestamp_basis=utc_corrected`. Default `broker_local` is byte-identical legacy.
- **Relationships:** `mt5_candle_fetcher` labeling; `feature_pipeline` session path.
- **Chapter:** Ch.07; F-066.

5. Semantic stack (states -> context -> shape -> CRT resolve -> model evidence)

`src/features/feature_states.py`

- **Group:** A * **Relevance:** LIVE-semantic
- **Purpose:** Layer 2 - feature VALUES -> declared semantic STATES (`market_crt_states` / ontology `feature_states`).
- **Chapter:** Ch.06-08 semantic pipeline.

`src/features/market_context.py`

- **Group:** A * **Relevance:** LIVE-semantic
- **Purpose:** Layer 4 - combine states into full market context ("what is happening").
- **Chapter:** Ch.06.

`src/features/market_shape.py`

- **Group:** A * **Relevance:** LIVE-semantic
- **Purpose:** Layer 5 - assign recurring shape identity (`configs/formulas/market_shapes.yaml` class).
- **Chapter:** Ch.06 / Ch.09 adjacent.

`src/features/crt_state_resolver.py`

- **Group:** A * **Relevance:** LIVE-semantic / RESEARCH-parity
- **Purpose:** Resolve 9 CRT states from declared feature states (declarative path). Semantic parity with `crt_engine_v2` is incomplete by construction (F-069) - not a drop-in replacement for the engine.
- **Chapter:** Ch.08; F-069.

`src/features/model_evidence.py`

- **Group:** A * **Relevance:** LIVE-semantic
- **Purpose:** Layer 7 - record what each model says about the bar (testimony), not what the market is.
- **Chapter:** Ch.06 semantic stack.

6. Dataset, validation, drift

`src/features/dataset_builder.py`

- **Group:** A * **Relevance:** TRAIN/RESEARCH
- **Purpose:** Extract feature vectors / BitNet slices from fusion logs for training datasets.
- **Chapter:** Ch.07; training E4/E6.

`src/features/dataset_validator.py`

- **Group:** A * **Relevance:** TRAIN/RESEARCH
- **Purpose:** Validate fusion trade logs before dataset build or model training (entropy, balance, min-n).
- **Chapter:** Ch.16 training quality gates.

src/features/feature_monitor.py

- **Group:** A * **Relevance:** LIVE-ops
- **Purpose:** Rolling feature distribution + drift (Z-scores); wired on TRADE_OPENED in backtest (CLAUDE Phase 2).
- **Chapter:** Ch.07; FeatureMonitor in backtest_v2.

src/features/__init__.py

- **Group:** A * **Relevance:** LIVE
- **Purpose:** Package exports / marker.

Config keys (feature surface)

Config area	Examples	Effect
feature_pipeline	rsi_period, atr_period, ema_*, macd_*, normalization_basis, session_timestamp_basis	Periods + F-061/F-066 gates
Ontology YAML	configs/formulas/market_ontology.yaml	Entity identities / lifecycle
Shapes YAML	configs/formulas/market_shapes.yaml	Shape declarations
CRT states YAML	market_crt_states (semantic)	State predicates for resolver

Tests & lint (pointers)

Surface	Role
tests/test_feature_pipeline.py, tests/test_candle_math.py	Pipeline / geometry
scripts/analysis/feature_math_lint.py	Ownership lint - no new local formula math
E3 feature_dag_*, feature_surface_query	Certification / query API
tests/test_schema_contracts.py	Schema pins

Danger flags

Flag	Meaning
NO_LOCAL_FORMULA_MATH	New quantities register in ontology -> registry first
PIT_SENSITIVE	Swings/structure must stay causal (F-051)
DIM_PIN	Schema hash load-bearing - dim change invalidates baselines

RESOLVER_!=_ENGINE	crt_state_resolver not byte-identical to crt_engine_v2 (F-069)
DOC_DRIFT	schemas.md may still say 38-dim - code is 39

E1b exit criterion

Criterion	Status
All 28 src/features/**/* .py files have purpose rows	Met
Registry authority chain documented	Met
Semantic layers L2/L4/L5/L7 + CRT resolver placed	Met
Linked to Ch.06-07 and F-ids	Met

Related: Ch.06 (../06-market-ontology.md) * Ch.07 (../07-feature-pipeline.md) * [E1](#) * [E3 feature DAG](#) *
Machine twin: encyclopedia_rows.jsonl

Encyclopedia E2 - Research Utilities Index

Phase: E2
Status: DONE (category + package + driver index - not 269 prose essays)
Date: 2026-08-07
Parent: [Chapter 24](#) * [Encyclopedia index](#)
Architecture narrative: Chapter 19 (`../19-research-programs.md`) * Chapter 20 (`../20-research-platform.md`)

Scope

Tree	.py files	Role
src/research/	143	Kernels, hypotheses, measurement, adapters - importable library
scripts/research/	126	Thin CLI drivers that write results/research/**
Total E2 surface	269	Research-only authority unless a finding says otherwise

E2 design rule (from charter): index by **category** + **entry point** + **program link**, not a narrative chapter per file. Deep math stays in pre-registrations + findings.

Group: mostly **A-adjacent research** for kernels used by Programs 1-9; **D** for one-off scripts.
Relevance: almost all **RESEARCH** - no production fusion weight without \$6.5 Authority Ladder.

How to use this index

1. Start from a **Program** (table below) -> kernel + driver + artifact.
2. Or start from a `src/research/` **package** -> file list.
3. Or start from a **script cluster** -> which CLIs exist.
4. Confirm claims in `docs/current-findings.md` + `results/research/**`.

Rule: a claim without an artifact path (or finding Evidence) is a hypothesis, not a result.

1. Program -> code -> driver -> artifact (primary map)

Prog	Question (short)	Primary src/research	Typical scripts/research drivers	Typical artifacts
1	Directional edge on crypto majors?	qualification.py, hypotheses/*, adapters/spine_signal_source.py, controls/*, costs.py	qualify_majors.py	results/research/qualification
1b	Conditional pockets?	conditional_entropy	phase_b_conditional	results/research/phase_b/
1c	RETEST selection skill?	selection_effect.py	phase_s_selection_eff	results/research/phase_s/

1d	Exit/cost grid?	exit_grid.py, costs.py	phase_d_exit_grid.py	results/research/phase_d/
2	Sweep->disp->retest asymmetry?	structural_asymmetry.py, adapters/structural_event_source.py	phase_e_structural_asymmetry.py	results/research/phase_e/
3	H1/H4 rescue?	resample.py, adapters + qualification	build_resampled_data.py, qualify_h1f.py	results/research/qualification/
P&F / IC	Interpreter edge?	src/interpreters/* + qualification.py	qualify_interpreter.py	Interpreter shadow results
4	Vol-regime level?	regime_conditioning.py	qualify_regime_conditioning.py	results/research/regime/
4b	Markov transition?	regime observers + conditioning	qualify_regime_transition.py	regime transition artifacts
4d / transitions	Compression->expansion info?	candle_state/*, hypotheses/compression.py	transition_information.py, qualify_transitions.py	results/research/candle_state/
5	Cross-sectional RV?	cross_sectional.py	qualify_cross_sectional.py	cross_sectional/*.json
6 / 6b	Carry signal / harvest?	cross_sectional.py + carry configs	qualify_carry.py, qualify_harvest.py	carry/*.json, harvest/*.json
8	Weekly sweep?	weekly_sweep/weekly_sweep.py, hypotheses/weekly_sweep_reversal.py	qualify_weekly_sweep.py	results/research/weekly_sweep/
9	M5 multi-TF + OCO straddle?	candle_state/m5_incremental.py, mtf_conjunction.py, hypotheses/compression.py, resample.py	m5_mtf_information.py, qualify_m5_straddle.py, verify_m5_resample_parity.py	results/research/m5_mtf/
FX gen.	Crypto null on FX?	same M4 stack	qualify_fx_metals.py, qualify_xauusd.py	FX qualification trees
Zone audits	ZoneGate labels / inertness?	zone_mapping/*, zone_label_audit.py	zone_label_audit.py, diagnose_zone_inertness.py, discover_zones.py, ...	zone_* under results
RR clean labels	Honest RR labels?	clean_labels/*, measurement/forward.py	rr_l3_label_generator.py, rr_shadow_value.py, rr_kill_test_clean_l3.py	results/rr_research/ etc.
MSIP hyps	State vector info?	(shadow in src/msip - E1)	run_h_msip_001.py, run_h_msip_002.py	results/research/h_msip_001/ h_msip_002/
Gate measure	Fusion gate veto rate?	spine adapters	gate_measurement_measurement.py	gate_measurement_*

Shared **M4 gate** implementation center of gravity: `src/research/qualification.py` (+ `costs.py`, `measurement/*`, `controls/*`).

2. Shared research spine (load-bearing root modules)

These top-level modules under `src/research/` are reused across many programs:

File	Purpose	Relevance
qualification.py	M4 QualificationGate - absolute expectancy, controls, permutations, PROMOTE/REJECT	LIVE-research
costs.py	Round-trip cost model (e.g. 12 bps standard)	LIVE-research
exit_grid.py	SLxTP grid evaluations	P1d
cross_sectional.py	Panel / relative-value kernel	P5/P6/P6b
conditional_entropy_grid.py	Entropy + economics grid	P1b
selection_effect.py	Selected vs rejected retest decomposition	P1c
structural_asymmetry.py	Funnel asymmetry vs controls	P2
regime_conditioning.py	Regime x consumer matrix	P4/4b
resample.py	Deterministic calendar OHLCV resampler	P3, P9
experiment_spec.py	Experiment specification objects	ERP hygiene
config.py	Research config load helpers	all
contracts.py	Research contract types	ERP
runner.py / cli.py	Generic research runner / CLI glue	ops
registry.py	Research registry helpers	ERP
provenance.py	Run provenance stamps	hygiene
forensics.py	Trade/path forensics helpers	diagnostics
goal_alignment.py	Goal-layer alignment checks	Ch.01 link
process_diagnostics.py / process_characterization.py	Process / entropy diagnostics	P1b sanity
indicators.py	Research-side indicators (not production feature authority)	support
band_validation.py	Band / threshold validation helpers	FM/feature research
shape_statistics.py	Shape library statistics	IC tracks
mt5_cost_calibration.py	FX/metals cost calibration	F-035 class
xau_metals_protocol.py	XAU metals research protocol	metals track
zone_label_audit.py	Zone label honesty / F-041B class	ZoneGate
__init__.py	Package marker	-

Authority: research kernels must not silently become production fusion weights. Production path remains EngineRunner spine (E1).

3. src/research/ package encyclopedia

3.1 adapters/ (4)

File	Purpose
spine_signal_source.py	Research adapter: CRT spine entries as signal source for M4
structural_event_source.py	Sweep/disp/retest structural events (Program 2)
shape_signal_source.py	Shape/interpreter signal source adapter
__init__.py	Package exports

3.2 controls/ (3)

File	Purpose
random_baseline.py	Random-uniform / shuffle controls
always_long.py	Long-only / market control
__init__.py	Exports

Every economic PROMOTE path is expected to beat appropriate controls (program-specific).

3.3 hypotheses/ (8) - Stage-2 economic consumers

File	Purpose	Programs
expansion_breakout.py	Directional expansion breakout consumer	P1, P3, P4 toys
mean_reversion.py	Mean-reversion consumer	P1, P4
compression_breakout.py	Compression->expansion directional consumer	P4d Stage-2
compression_box_straddle.py	Sole P9 Stage-2 OCO straddle consumer	P9
weekly_sweep_reversal.py	Weekly sweep reversal consumer	P8
spine_hypothesis.py	Spine entry hypothesis wrapper	P1 spine arm
market_shape_hypothesis.py	Market-shape hypothesis	IC / shape
__init__.py	Exports	

3.4 candle_state/ (7) - multi-TF state & Program 9 Stage-1

File	Purpose
encoder.py	Candle-state encoding
mtf_conjunction.py	M15^H1^H4 (and extensions) conjunction builder
transition_target.py	Forward vol/range/transition targets (Program 4 kernels)
m5_incremental.py	M5-base keys + incremental information gate (P9 Stage-1)
info_robustness.py	Information robustness helpers
reporting.py	Stage-1 reporting
__init__.py	Exports

3.5 measurement/ (4) - honest outcomes

File	Purpose
forward_walk.py	Intrabar-aware forward walk labels (anti-F-022 contamination)
metrics.py	Research metrics (E, PF, ...)
bootstrap.py	Bootstrap CIs
__init__.py	Exports

MC path: schema in governance; this package is scaffolding toward sealed mc- * (still OPEN - Ch.20).

3.6 clean_labels/ (3)

File	Purpose
builder.py	Build clean labels from geometry + forward_walk
protocol.py	Clean-label protocol freeze
__init__.py	Exports

3.7 weekly_sweep/ (2)

File	Purpose
weekly_range.py	Calendar-locked weekly range + sweep geometry (P8)
__init__.py	Exports

3.8 zone_mapping/ (13)

Zone census, historical mapping, rare-zone evals, Gaussian-family shadows, CRTxzone crosstabs - research audits of ZoneGate / geometry (F-036, F-041 class). Files:

boundary_hypothesis_eval.py, build_corpus_zone_map.py, collect_trade_opened_features.py, crt_zone_crosstab.py, displacement_zone_event_study.py, gaussian_delta_gap_eval.py, gaussian_family_shadow_eval.py, historical_zone_mapper.py, rare_zone_context_filter_eval.py,

`rare_zone_detection_eval.py`, `rare_zone_fa_characterization.py`, `zone_census.py`, `__init__.py`.

3.9 episodes/ (13+)

Episode store / schema / tensors / projectors for trajectory research (not the live single-candle spine).

Core: `builder.py`, `events.py`, `flat.py`, `policy.py`, `protocol.py`, `query.py`, `schema.py`, `store.py`, `tensors.py`, `projectors/detection.py`, `projectors/spine.py`, `__init__.py`.

3.10 ic002_entry_evolution/ (5)

Trajectory build + separation eval for entry-evolution interpreter candidates: `build_trajectories.py`, `evaluate_separation.py`, `io_util.py`, `schema.py`, `__init__.py`.

3.11 ic003_shapes/ (3)

Shape library: `build_library.py`, `schema.py`, `__init__.py`.

3.12 ic003b_sequence_geometry/ (8)

Sequence geometry / DTW clustering: `build_all.py`, `cluster_dtw.py`, `cluster_euclid.py`, `continuum.py`, `dtw.py`, `schema.py`, `summarize.py`, `__init__.py`.

3.13 secondlow_v1/ (5)

Second-low detector track: `corpus.py`, `depth_metrics.py`, `detector.py`, `regime_metrics.py`, `__init__.py`.

3.14 path/ (2)

`ambiguity_census.py` - path ambiguity census (intrabar / touch-order class issues).

3.15 envelope_offline/ (3)

Offline envelope train/shadow: `train.py`, `shadow.py`, `__init__.py` (TradeNet/envelope research - not live fusion authority).

3.16 model_runners/ (23)

Generic offline model runner stack: `runner.py`, `envelope.py`, `contracts.py`, `require_config.py`, `schema_resolver.py`, `stats.py`, `substrate.py`, plus `adapters/*` for model families. **Research execution substrate**, not EngineRunner.

3.17 synthetic/ (10)

Synthetic story ontology for controlled experiments: `ontology.py`, `story_builder.py`, `story_registry.py`, `story_spec.py`, `stories/*`.

3.18 Root-only modules

Listed in §2 shared spine table (27 root `.py` including package init).

4. scripts/research/ - driver clusters (126 files)

Thin wrappers: **prefer** importing `src/research` kernels; scripts should not reinvent gate math.

4.1 Qualify / M4 gate (16) - primary economic runners

`qualify_majors.py`, `qualify_htf.py`, `qualify_fx_metals.py`, `qualify_xauusd.py`, `qualify_transitions.py`,
`qualify_regime_conditioning.py`, `qualify_regime_transition.py`, `qualify_cross_sectional.py`,
`qualify_carry.py`, `qualify_harvest.py`, `qualify_weekly_sweep.py`, `qualify_m5_straddle.py`,
`qualify_interpreter.py`, `qualify_shape_xauusd.py`, `qualify_zone_topk.py`, `xauusd_gaussian_m4_qualify.py`.

When to run: only against a **pre-registration** and frozen config; write under `results/research/`.

4.2 Phase / Program 1 sub-drivers (4)

`phase_b_conditional_entropy.py`, `phase_d_exit_grid.py`, `phase_e_structural_asymmetry.py`,
`phase_s_selection_effect.py`.

4.3 Regime / transition information (1+)

`transition_information.py` (Stage-1 info); economic twin often `qualify_transitions.py` / regime qualify scripts.

4.4 Program 9 M5 (2+)

`m5_mtf_information.py` (Stage-1), `verify_m5_resample_parity.py`, `qualify_m5_straddle.py` (Stage-2, listed under qualify).

4.5 Zone / ZoneGate research (14)

`discover_zones.py`, `zone_label_audit.py`, `diagnose_zone_inertness.py`, `zone_gate_threshold_sweep.py`,
`ablate_zone_thr_xauusd_fusion.py`, `convert_zones_v1_to_gaussian.py`, `build_zone_census.py`,
`build crt_zone_crosstab.py`, `build_displacement_zone_event_study.py`, rare-zone builders,
`trace_zone_gate_xauusd.py`, `zone_x_o4_gap_study.py`, ...

4.6 RR / clean labels (11)

`rr_l3_label_generation.py`, `rr_l2_feature_truth.py`, `rr_l4_research_execution.py`, `rr_shadow_value.py`,
`rr_kill_test_clean_l3.py`, `build_clean_labels_tn_env.py`, `build_rr_dataset_from_clean_labels.py`,
`analyze_clean_labels_tn_env.py`, `h_rr_threshold_001.py`, `gaussian_rr_scatter.py`,
`bitnet_population_label_audit.py`.

4.7 BitNet / Gaussian / TradeNet diagnostics (8)

`bitnet_r25_kill_test.py`, `bitnet_shadow_diagnostic.py`, `diagnose_gaussian_pivotality.py`, Gaussian
family/delta builders, XAUUSD Gaussian econ ledgers, TradeNet report patches.

4.8 CRT parity / resolver (7)

`crt_parity_*.py`, `crt_state_confusion_matrix.py`, `validate_crt_state_resolver.py`,
`crt_range_rebuild_probe.py`, `crt_resolver_economic_comparison.py`.

4.9 IC / shapes (6)

`ic002_build_trajectories.py`, `ic002_evaluate.py`, `ic003_build_shape_library.py`, `ic003b_build.py`,
`ic003b_derive_information.py`, `shape_statistics_survey.py`.

4.10 Episodes / corpus / path (13)

`build_episodes.py`, `build_trace_corpus.py`, `analyze_trace_corpus.py`, `certify_xauusd_corpus.py`, path ambiguity
/ knn / geometry tracers, enrich helpers.

4.11 XAU / metals / FX cost (8)

`run_xau_metals_protocol_v1.py`, `xauusd_mt5_cost_calibration.py`, `rerun_xau_metals_m4.py`, MT5 CRT runners, library build helpers.

4.12 MSIP hypothesis runners (2)

`run_h_msip_001.py`, `run_h_msip_002.py` - research-only; no CRT production authority.

4.13 Gate measurement (2)

`gate_measurement_m_gate_01.py` (F-070 class), `gate_o_nonlinear_probe.py`.

4.14 Private / underscore helpers (4+)

`_verify_outputs.py`, `_watch_h_rr.py`, `_fix_eval_auc.py`, `_show_top_decile_n9.py`, `_corpus_inspect.py`, ... - operator helpers, not programs.

4.15 Uncategorized residual (~28)

Includes forensics (`bnbusdt_forensics.py`), attribution studies, envelope train/shadow, story library, promotion dry-run, opportunity scanner, dimensional-mix diagnostics, FM030 band validation, model offline runners, etc. **Treat as ops/research tools:** read the module docstring before running; assume **write to results/** unless proven read-only.

5. Configs, pre-regs, artifacts (not code but E2 surface)

Kind	Path
Research configs	<code>configs/research/research_config*.json</code> , <code>configs/research/experiments/</code>
Measurement profiles	<code>configs/research/measurement_contracts/*.v1.json</code>
Pre-registrations	<code>docs/research/preregistration-*.md</code> , <code>docs/research-readiness/program-*.preregistration.md</code>
Findings	<code>docs/current-findings.md</code>
Family registry	<code>docs/governance/research_family_registry.json</code>
Outputs	<code>results/research/**</code>

6. Danger / authority flags

Flag	Meaning
<code>RESEARCH_ONLY</code>	Default for all of E2 - no fusion/production weight
<code>WRITES_ARTIFACTS</code>	Most <code>scripts/research/*</code> write under <code>results/</code>
<code>LABEL_SENSITIVE</code>	RR/zone label scripts - F-022 contamination class; prefer <code>forward_walk</code> / <code>clean_labels</code>

COST_MODEL	Changing bps or exit model invalidates cross-program comparison (Measurement Contract gap)
PRE_REG_REQUIRED	New economic claims need a frozen pre-registration first
OBSERVE_ONLY	Parity/trace/counter scripts must not change spine math

E2 exit criterion

Criterion	Status
Every src/research subpackage has a purpose row	Met (§3)
Shared root kernels listed	Met (§2)
Programs 1-9 mapped to kernels/drivers/artifacts	Met (§1)
All 126 scripts classified into clusters (or residual)	Met (§4)
Per-file essay for all 269	Out of scope (charter: index, not encyclopedia essays)

Next

- **E3** - Governance tooling (src/governance/, scripts/governance/, scripts/analysis/).
 - Optional deep-dives: one program appendix (e.g. P9 only) if Stage-1/2 re-runs are planned.
-

Related: Ch.19 ([../19-research-programs.md](#)) * Ch.20 ([../20-research-platform.md](#)) * [E1 spine](#) * [Ch.24](#)

Encyclopedia E3 - Governance Tooling

Phase: E3

Status: DONE (library + CLI index - not every analysis probe essay)

Date: 2026-08-07

Parent: [Chapter 24](#) * [Encyclopedia index](#)

Architecture narrative: Chapter 16 ([../16-config-first-and-promotion.md](#)) * Chapter 17

([../17-truth-maintenance.md](#)) * Chapter 18 ([../18-field-guide-governance.md](#))

Scope

Tree	.py files	Role
src/governance/	19	Promotion, registries, validators, SITS cores - importable
scripts/governance/	31	Construction protocol, seeds, queries, feature-DAG, promote CLIs
scripts/analysis/	115	Census, probes, generators, instrument forensics (OBSERVE-heavy)
scripts/maintenance/	9	Hash, session-log rotation, hygiene (adjacent ops)
Total E3 surface	~174	Governance / truth / map tooling

E3 design rule: index by **job type** (promote, seed registry, certify features, generate maps, probe drift) - not 174 narratives. Deep doctrine stays in Ch.16-18 and [docs/governance/](#)*.

Group: governance libraries = **A-adjacent (ops authority)**; most analysis scripts = **D (operators)** or **OBSERVE_ONLY**.

Relevance: can **write** promotion logs, registries under [data/](#), governance artifacts - treat write paths as gated.

How to use this index

If you need to...	Start here
Promote a config	PromotionManager + scripts/governance/promote_v2.py * Ch.16
Enforce construction protocol	scripts/governance/construction_protocol.py
Query hypotheses / frameworks / scripts	query_hypotheses.py / query_registry.py / query_scripts.py
Seed registries	seed_*_registry.py (PRIMARY in seeds, GENERATED in data/)
Feature certification / DAG	feature_dag_certify.py , feature_certification_state.py , feature_surface_query.py
Findings machine export	export_findings.py -> data/findings.jsonl

Config reachability / behavior census	scripts/analysis/config_reachability.py, behavior_census.py
Regenerate architecture maps	gen_code_map.py, gen_pyan.py, generate_cli_matrix.py, ...

1. Job map (primary navigation)

Job	Library (src/governance)	CLI / script	Writes	Chapter
Promote production config	<code>promotion_manager.py</code>	<code>scripts/governance/promote.py</code>	<code>configs/production/*</code> , <code>promotion_log.jsonl</code>	Ch.16
Shadow promotion gate	<code>shadow_promotion_gate.py</code>	<code>scripts/governance/v2/manager /</code> <code>(dedicated flows)</code>	shadow reports	Ch.16
Config integrity check	<code>config_integrity.py</code>	analysis / preflight	usually none	Ch.16; F-006 class
Portfolio validation	<code>portfolio_validation.py</code>	multi-market validation paths	validation reports	Ch.16
Multi-strategy validation	<code>multi_strategy_validation.py</code> , <code>strategy_backtest.py</code>	<code>scripts/governance/v2 /</code> validators	ValidationReport	Ch.16
Governance loop orchestrator	<code>orchestrator.py</code>	control-plane / CLI wrappers	audit trails	Ch.16 adjacent
Findings -> JSONL	<code>findings_export.py</code>	<code>export_findings.py</code>	<code>data/findings.jsonl</code> (GENERATED)	Ch.17
Hypothesis registry	<code>hypothesis_registry.py</code>	<code>seed_hypothesis_registry.py</code> , <code>query_hypotheses.py</code>	<code>data/hypothesis_registry.jsonl</code>	Ch.17/JSONL
Framework registry	<code>framework_registry.py</code>	<code>seed_framework_registry.py</code> , <code>query_registry.py</code> , <code>update_registry.py</code>	<code>data/framework_registry.jsonl</code>	Ch.17/JSONL
Script registry (SITS)	<code>script_registry.py</code> , <code>script_census.py</code> , <code>script_seed.py</code>	<code>seed_script_registry.py</code> , <code>query_scripts.py</code> , <code>scripts/analysis/script_census.py</code>	<code>stubs +</code> <code>data/script_registry.jsonl</code>	CLAUDE §3.1
Module attribution census	<code>module_census.py</code> , <code>module_attribution.py</code>	module census CLI under analysis	module registry artifacts	Ch.04 adjacent
Construction protocol	-	<code>construction_protocol.py</code>	completion validation	Gate 6 protocol
Feature DAG certify	-	<code>feature_dag_certify.py</code> , <code>feature_certification_state.py</code>	certification ledger	Ch.06-07 / F-054
Feature surface query	-	<code>feature_surface_query.py</code>	read-only	Ch.07 query surface
BitNet governance executor	<code>bitnet_governance_executor.py</code>	BitNet governance flows	gated	Ch.10 BitNet
Meta governor executor	(if present as meta path)	meta-governor flows	gated	Ch.16

Expansion integration	expansion_integration_ops	expansion ops	research/ops	sidecar
Reflection buffer	reflection_buffer_advanced	advanced reflection	sidecar	low priority
Runtime import boundary scan	-	scan_runtime_boundary_report	report	hygiene
Model path literal scan	-	scan_model_paths_literal.py	report	ModelPaths boundary
Three-authority surplus census	-	three_authority_surplus_census.py	license artifacts	Ch.17 / WHO-WHAT-HOW
CRT config construction census	-	crt_config_construction_census(F-057 class)	license(F-057 class)	Ch.13
Validation access pack	-	validation_access_cli	results/validation_access	research/** access
Session log / hash hygiene	-	scripts/maintenance/*	logs, hashes	CLAUDE §6

2. src/governance/ - library encyclopedia (19 files)

Promotion & validation (LIVE ops)

`promotion_manager.py`

- **Relevance:** LIVE * **Purpose:** Only path to production promotion - requires APPROVE ValidationReport, SHA-256, append-only promotion_log.jsonl.
- **Entry:** python .../promotion_manager.py Or promote_v2.py; never bypass.
- **Chapter:** Ch.16.

`shadow_promotion_gate.py`

- **Relevance:** LIVE * **Purpose:** Two-gate shadow promotion before full promote.
- **Chapter:** Ch.16.

`config_integrity.py`

- **Relevance:** LIVE (often under-consumed) * **Purpose:** Config lineage / integrity guards (F-006: real check historically orphaned from hot path - still hygiene-critical).
- **Chapter:** Ch.16; findings F-006.

`portfolio_validation.py`

- **Relevance:** LIVE-research/ops * **Purpose:** Multi-market edge universality validation for portfolios.
- **Chapter:** Ch.16 adjacent; F-013 class (allocator orphaned from live single-candle spine).

`multi_strategy_validator.py` / `strategy_backtest.py`

- **Relevance:** PARTIAL * **Purpose:** ValidationReport across strategy modules + per-strategy backtest measurement.
- **Note:** Strategy modules largely DORMANT (Group C) - validators exist for multi-strategy governance path.

- **Chapter:** Ch.16; E5 for strategies themselves.

`orchestrator.py`

- **Relevance:** LIVE-ops * **Purpose:** GovernanceOrchestrator - multi-step governance loop coordination.
- **Chapter:** Ch.16.

Registries & generated truth (LIVE meta)

`findings_export.py`

- **Relevance:** LIVE * **Purpose:** Parse `docs/current-findings.md` -> `data/findings.jsonl` (GENERATED; never hand-edit).
- **CLI:** `scripts/governance/export_findings.py`
- **Chapter:** Ch.17.

`hypothesis_registry.py`

- **Relevance:** LIVE * **Purpose:** H-id registry schema/validation; authority locked to research.
- **Seed/query:** `seed_hypothesis_registry.py`, `query_hypotheses.py`
- **Chapter:** Ch.17 / E2.

`framework_registry.py`

- **Relevance:** LIVE * **Purpose:** Architecture component map (`kernel/domain/style/...`) as JSONL registry.
- **Seed/query/update:** `seed_framework_registry.py`, `query_registry.py`, `update_registry.py`
- **Chapter:** Ch.17.

`script_registry.py` / `script_census.py` / `script_seed.py`

- **Relevance:** LIVE * **Purpose:** SITS - discover scripts, merge stubs/overlays, materialize `data/script_registry.jsonl`.
- **CLI:** `seed_script_registry.py`, `query_scripts.py`, `scripts/analysis/script_census.py` (+ matrix generators).
- **Chapter:** CLAUDE.md §3.1; conventions.

`module_census.py` / `module_attribution.py`

- **Relevance:** LIVE * **Purpose:** Same idea as SITS for every `src/**/*.py` - ownership / attribution surface.
- **Chapter:** Ch.04 adjacent.

Specialist / sidecar

File	Relevance	Purpose
<code>bitnet_governance_executor.py</code>	SPECIALIST	BitNet-specific governance execution path
<code>expansion_integration.py</code>	SIDECAR	Wire expansion engine into governance loops
<code>reflection_buffer_advanced.py</code>	SIDECAR	Advanced reflection buffer for governance agents
<code>__init__.py</code>	LIVE	Package marker

3. scripts/governance/ - CLI encyclopedia (31 files)

3.1 Construction & promotion

Script	Purpose	Danger
construction_protocol.py	Gate-6 BUILD_IMPACT_MANIFEST / validate-completion / check	WRITE policy enforcement
promote_v2.py	Thin multi-strategy promote CLI	WRITES production configs
crt_config_construction_census.py	ConfigBuilder / CRTConfig construction sites (F-057 observe)	READ-ONLY census
behavioral_constant_authority_trace.py	Behavioral constants on decision spine	READ-ONLY

3.2 Registry seeds & queries

Script	Purpose
seed_hypothesis_registry.py	Seed H-ids from curated ledger
seed_framework_registry.py	Seed framework registry
seed_script_registry.py	SITS seed + PRIMARY overlays
export_findings.py	Regenerate findings JSONL
query_hypotheses.py / query_registry.py / query_scripts.py	Read-only queries
update_registry.py	Append-only framework registry update
framework_registry_report.py / framework_gap_audit.py	Markdown reports from registry

3.3 Feature DAG / FM governance

Script	Purpose
feature_certification_state.py	Per-feature certification ledger seed/resolver
feature_dag_certify.py	Certify / promote / STALE cascade mutations
feature_surface_query.py	Read-only join API over feature surface artifacts
build_fm_ownership_matrix.py	FM ownership / consumer matrix generator

3.4 Authority censuses & boundary scans

Script	Purpose
three_authority_surplus_census.py	WHO/WHAT/HOW declaration surplus
who_numeric_dependency_census.py	WHO numeric threshold dependency census
scan_runtime_boundary.py	Kernel ↔ research import direction + promotion boundary

scan_model_paths_literals.py	Unauthorized models/ path literals
build_g001_consumer_attribution.py	G001 consumer attribution package

3.5 Decision packages / MSIP / RR freeze / zone remap

Script	Purpose	Class
build crt architecture adjudication	CRT architecture adjudication package	OBSERVE package
build msip shadow design contract	MSIP shadow design freeze package	design only
build msip1 verification package.py	MSIP-1 verification pack	OBSERVE
patch msip shadow design consistency	Consistency patch for MSIP design pack	OBSERVE
rr_l1_freeze_certificate.py	RR L1 freeze certificate hash/assert	certificate
remap_zone_registry_v4.py	Zone registry schema v3->v4 remap	WRITES model JSON (one-shot)
validation_access_cli.py	VA-XAUUSD-M15 dual-surface access ladder	writes results packs
_build_doc_tracking_index.py	DOC_TRACKING_INDEX.xlsx metadata inventory	Excel write

4. scripts/analysis/ - census & probe clusters (115 files)

Default stance: OBSERVE_ONLY / diagnostic. Prefer governance feature-DAG tools for **mutations**; analysis scripts for **measurement**.

Cluster	~Count	Examples	Use when
Feature DAG / FM certification	15	feature_dag_layers.py, feature_math_lint.py, b2a_feature_candidate_certification.py, FM drift/decision-flip probes	Feature surface hygiene, F-053/F-054 class
CRT / state / parity	14	funnel diagnostics, guard ablations, transition audits, soft-conf / shadow probes	CRT CLOSED boundary forensics
RR / Gaussian / Zone / BitNet probes	10	rr_confidence_probe.py, zone parity, gaussian/bitnet runners	Engine AUDITED lineage evidence
XAUUSD / instrument forensics	9	trade anatomy, timestamp gaps, phase1 validation	Instrument-specific audits
Code map / wiring / matrices	7	gen_code_map.py, gen_pyan.py, generate_cli_matrix.py, generate_script_matrix.py, script_census.py	Regenerate architecture/CLI maps
Session / clock / normalization	5	session certification, cost audit, override scoping	F-066 / session policy class

Config / behavior census	2	config_reachability.py, behavior_census.py	Config-first maturity (\$6.5)
Blind label / epistemic	2	blind_label_sample/score	Epistemic / labeling studies
Geometry census	1+	geometry_census.py	Geometry family registry inputs
Other / residual	~50	pattern library, consensus sweeps, log compress, early invalidation A/B, detection sweeps, ...	Read docstring; classify before reuse

Regenerate maps (safe, high-ROI):

```
scripts/analysis/gen_code_map.py scripts/analysis/gen_pyan.py
scripts/analysis/generate_cli_matrix.py scripts/analysis/generate_script_matrix.py
scripts/analysis/script_census.py # often with --write-stubs for SITS
```

5. scripts/maintenance/ - adjacent hygiene (9 files)

Script	Purpose
_compute_hash.py	Recompute production config hash after params edits
rotate_session_log.py	Rotate assistant_project.md / workflow log
check_session_log_commit.py	Commit-hook session log presence
check_governance_invariants.py	Governance invariant checker
check_consolidation_due.py	Consolidation due probe
cleanup_logs.py / reorganize_results.py	Log/results hygiene
fix_bom.py	BOM fixes
_promote_v4_bnb_cutover.py	Historical cutover helper (danger: version-specific)

6. Evidence ledger home (docs, not scripts)

E3 tools **read/write into** the evidence directory mapped in Ch.18:

Cluster	Examples
Closure reports	crt_closure_report.md, feature surface closures
Lineage audits	*_lineage_audit.md
Certification ledgers	feature_certification_ledger.jsonl, DAG LATEST JSON
Measurement contract	MEASUREMENT_CONTRACT.md (still OPEN seals)
Construction protocol	REPOSITORY_CONSTRUCTION_PROTOCOL.md, change_contracts.json

MIAR	MODEL_INTENT_AUTHORITY_REGISTER.md
------	------------------------------------

7. Danger / authority flags

Flag	Meaning
PROMOTION_WRITE	Can change active production config - human confirm required
GENERATED_ONLY	data/*.jsonl - regenerate from seeds/docs; never hand-edit as truth
APPEND_ONLY	Registries / promotion_log / certification ledger - no silent rewrite
OBSERVE_ONLY	Analysis probes must not change production behavior
ONE_SHOT_MUTATION	e.g. zone registry remap - archive + hash after
ORPHANED_CHECK	e.g. config_integrity may not gate runtime - still run in hygiene

E3 exit criterion

Criterion	Status
Every src/governance module has a purpose row	Met (§2)
All scripts/governance listed by job cluster	Met (§3)
scripts/analysis clustered with regenerate shortlist	Met (§4)
Maintenance adjacency noted	Met (§5)
Per-file essay for all 115 analysis scripts	Out of scope (cluster index)

Next

- **E4** - Group B sidecars (bitnet, utils, training, portfolio, replay, ...)
- **E5** - Group C dormant one-liners
- **E6** - Remaining non-research / non-governance scripts catalog

Related: Ch.16 (../16-config-first-and-promotion.md) * Ch.17 (../17-truth-maintenance.md) * Ch.18 (../18-field-guide-governance.md) * [E1](#) * [E2](#) * [Ch.24](#)

Encyclopedia E4 - Sidecar / Support Modules (Group B)

Phase: E4
Status: DONE (package + file map for non-spine support surfaces)
Date: 2026-08-07
Parent: [Chapter 24](#) * [Encyclopedia index](#)

Scope

Group B packages **support** the system but are **not required** to understand the candle->order spine on day one. They are not all equally “dead” - some are LIVE data/ops helpers; many are INERT, UNWIRED, RESEARCH, or OBSERVE_ONLY on the active production config.

Package	.py files	Default posture
src/bitnet/	23	Built; inert when use_bitnet: (F-004/F-055)
src/utils/	17	LIVE helpers (logging, JSONL, manifests)
src/retrieval/	9	Enterprise RAG - optional tooling
src/training/	8	Offline train pipelines; TradeNet built/unwired (F-005)
src/expansion/	6	Controlled frequency expansion - ops/research
src/portfolio/	6	Allocator built; live spine still single-candle (F-013)

src/replay/	6	Replay memory / timing - sidecar (F-012 class)
src/analytics/	5	Offline metrics / clustering
src/regime/	4	Regime clustering / routing - research-adja
src/msip/	6	Shadow market state - OBSERVE_O (Ch.18)
src/multi_llm/	5	Multi-LLM ops tooling (also root multi_llm/)
src/events/	2	Event fabric
src/search/	1	Regime weight searcher
src/monitoring/	2	Health endpoint (borderline dormant/ops)
Total	~100	

Not in E4: Group C dormant strategies/scanner/journal (-> **E5**); research/gov (E2/E3); spine (E1).

Rule: SIDECAR != deleted. Read when a task touches the package; do not promote into fusion without Authority Ladder evidence.

How to use

Need	Package
BitNet zone gate / low-bit models	bitnet/ + Ch.10

Logging / JSONL / run manifests	utils/
Offline train / TradeNet	training/
Portfolio multi-instrument	portfolio/
Replay memory / pattern timing	replay/
Offline performance metrics	analytics/
Shadow MSV	msip/
Multi-LLM turn ledger / packs	multi_llm/ + Ch.23
Codebase RAG	retrieval/
Parameter expansion	expansion/

1. src/bitnet/ - BitNet / zone intelligence stack (23)

Posture: Full stack **built**; production **INERT** while use_bitnet false on active patch.
Shadow A/B (F-055) did not earn activation authority.

File	Purpose
__init__.py	Package: BitNet parameter / zone discovery engine
bitnet_inference.py	Inference for dual JSON model schemas
bitnet_runner.py	Schema enforcement + fail-closed dimension checks
bitnet_registry.py	Version registry - Exists != Selected != Enabled
model_contract.py	Canonical serialization contract (bitnet_v3)
model_bundle.py	CONTRACT-C model.bundle write/validate
composition.py	Encoder -> Backbone -> Heads -> Adapter composition
encoders.py / backbones.py / heads.py / adapters.py	Spec hierarchy components
layers.py	Pure numeric BitLinear primitives
defaults.py	Versioned numeric defaults (Spec v1.2.1)
label_contracts.py	CONTRACT-C label definitions
contract_c_trainer.py	CONTRACT-C trainer for BitLinear residual family
runtime_types.py	Shared inference value types
zone_cosine_searcher.py	Cosine match to zone vectors
zone_validator.py	Min quality criteria for candidate zones
forward_tester.py	OOS validation for zone registry
stability_checker.py	Zone stability across temporal splits

population_label_audit.py	Read-only population/label balance audit
r25_kill_test.py	R2.5 kill-test harness
_smoke_test.py	Module smoke test

Related: src/engines/zone_gate_engine.py (E1), F-004/F-050/F-055, BitNet lineage audit.

Do not: enable on active config without ΔG001 + promotion path.

2. src/utils/ - shared helpers (17)

Posture: LIVE support - imported widely; not decision authority.

File	Purpose
logging_config.py	Centralized per-flow logging
console_safe.py	Windows/cp1252-safe console output
jsonl_writer.py	Canonical append-only JSONL helper
trade_logger.py	Trade event logging
llm_logger.py	Structured LLM session logger (UAT/SIT)
log_identity.py / log_index_writer.py	Log identity + index writers
engine_telemetry.py	Structured engine observability wrapper
integrity_events.py	Integrity-event telemetry spine
sweep_trace_logger.py	CRT sweep decision traces (sweep_trace.jsonl)
episode_summarizer.py	Episode summarization helpers
pattern_hasher.py	Stable pattern IDs for replay/memory
run_manifest.py	Provenance manifest for research runs (ERP H1/H2/H3)
validation_contract.py	Mechanical anti-hallucination checks over manifests
config_dumper.py	Dump resolved config to timestamped JSON
registry_refresh.py	mtime hot-reload tracker for model/config files
zone_schema_migrator.py	BitNet zone schema migration utility

3. src/training/ - offline training (8)

Posture: RESEARCH / OFFLINE. TradeNet **built but unwired** in fusion (F-005).

File	Purpose
train_pipeline.py	Main train pipeline (incl. Phase-5 calibration gate before register)
trainer.py	Trainer core loop

<code>evaluator.py</code>	Offline evaluation
<code>phase5_calibration.py</code>	Phase-5 calibration (note: historical gaussian artifact builder path)
<code>stage1_dataset_builder.py</code>	Stage-1 truth dataset for TradingLLM-class training
<code>trade_net_v2.py</code>	TradeNet v2 model definition / train surface
<code>training_trigger.py</code>	Programmatic gate to invoke auto-train
<code>bar_semantic_tracker.py</code>	Append-only bar-level semantic journal for train/eval

Related: E2 RR clean labels; `models/*` artifacts; never confuses train success with production authority.

4. `src/portfolio/` - multi-instrument allocation (6)

Posture: BUILT / ORPHANED from live single-candle spine (F-013). Research/ops use only until wired.

File	Purpose
<code>allocator.py</code>	PortfolioAllocator - multi-instrument capital allocation
<code>capital_policy.py</code>	Capital policy rules
<code>correlation_engine.py</code>	Cross-instrument correlation
<code>exposure_tracker.py</code>	Exposure tracking
<code>replay.py</code>	Read-only shadow replay of allocator over historical ledger
<code>__init__.py</code>	Package marker

5. `src/replay/` - institutional replay memory (6)

Posture: SIDECAR (F-012 class - zero spine consumption by default).

File	Purpose
<code>replay_memory_engine.py</code>	Persistent institutional memory layer
<code>replay_similarity_index.py</code>	Similarity search over replay vectors
<code>replay_drift_governor.py</code>	Contamination / staleness / cluster drift monitor
<code>timing_advisor.py</code>	Read API for Pattern Timing Library (measure-only Phase 1)
<code>timing_reconstructor.py</code>	Forward-walk timing augmentation of outcomes
<code>__init__.py</code>	Package marker

6. `src/analytics/` - offline analytics (5)

Posture: OFFLINE research/ops metrics (not live decision).

File	Purpose
metrics_oracle.py	Independent backtest metric recompute (trust layer)
performance.py	Win rate, expectancy, drawdown breakdowns
clustering.py	Cluster losing trades by condition
sl_tp_comparator.py	Dual SL/TP method comparator
__init__.py	Package exports

7. src/expansion/ - controlled frequency scaling (6)

Posture: OPS/RESEARCH - bounded parameter/frequency expansion; not free optimizer.

File	Purpose
expansion_engine.py	Expansion engine core
config_mutator.py	Config mutation helpers for expansion trials
evaluator.py	Expansion trial evaluation
policy_schema.py	Expansion policy schema
llm_pattern_extractor.py	Optional LLM pattern extraction for expansion
__init__.py	Package: controlled trade frequency scaling

8. src/regime/ - regime surfaces (4)

Posture: RESEARCH-adjacent (complements E2 regime_conditioning; not Ultron).

File	Purpose
regime_classifier.py	Regime classification
market_state_cluster_engine.py	Cluster-native market state classification
config_router.py	Regime -> config routing helper
__init__.py	Package marker

9. src/msip/ - Market State Interpretation Platform shadow (6)

Posture: OBSERVE_ONLY - never mutates CRT; never gates execution (Ch.18 brief).

File	Purpose
market_state_vector.py	MSV value object (schema pin)
interpretation_config.py	msip_shadow HOW config; off when absent/disabled

shadow_emitter.py	Pure build + append-only emit
disagreement.py	Shadow vs CRT disagreement taxonomy (telemetry)
isolation.py	Import/mutation isolation policy
__init__.py	Shadow package marker

Related: run_h_msip_001/002 (E2); no production cutover without explicit gates.

10. src/multi_llm/ - multi-LLM ops package (5)

Posture: OPS tooling for hand-operated multi-model workflow; **isolated from trading spine.**
(Repo root multi_llm/ holds protocol docs, ledger, roles - Ch.23.)

File	Purpose
turn_ledger.py	Append-only multi-LLM turn record
context_pack.py	Bounded upload-ready context pack for one task
discussion.py	Reconstruct / rewind discussion (view-only)
tokens.py	Heuristic token estimates for pack bounds
__init__.py	Coordination layer package

11. src/retrieval/ - codebase RAG (9)

Posture: OPTIONAL tooling (ChromaDB + embeddings). Not on candle->order path.

File	Purpose
retriever.py	High-level retrieval + context assembly
vector_store.py	ChromaDB hybrid search store
embedding.py	sentence-transformers embedding pipeline
chunking.py	Structure-aware semantic chunking
corpus.py	Repo document discovery by domain
config.py	Retrieval tunables
monitor.py	RAG metrics
claude_integration.py	Claude Code RAG-grounded task helpers
__init__.py	Enterprise RAG package

12. Small surfaces

src/events/ (2)

File	Purpose
event_fabric.py	Canonical Event Fabric for runtime eventing
__init__.py	Package marker

src/search/ (1)

File	Purpose
regime_weight_searcher.py	Search/sweep helpers for regime weights (research/ops)

src/monitoring/ (2)

File	Purpose
health_checker.py	Lightweight stdlib HTTP health on port 8788
__init__.py	Package marker

Note: Control plane dashboard is 8787 (E1); health checker is a separate lightweight surface.

Cross-cutting findings (why these stay sidecar)

Finding / class	Implication for E4
F-004 / F-055	BitNet hard gate inert / shadow not activated
F-005	TradeNet fusion slot unwired
F-012	Replay / cognitive / HMF sidecar-only
F-013	Portfolio allocator + scan loop orphaned from live single-candle spine
F-060	Live Gaussian channel near-constant - separate from bitnet/ package
Ch.18 MSIP	Shadow only; no CRT mutation

Danger flags

Flag	Meaning
INERT_UNTIL_CONFIG	Present in tree; inactive unless config enables (BitNet)
UNWIRED	Built but no production caller (TradeNet neural slot, portfolio live)
OBSERVE_ONLY	MSIP shadow, many replay/timing measure paths
OPTIONAL_DEPS	retrieval (ChromaDB, sentence-transformers) may be absent
NO_FUSION_AUTHORITY	None of E4 grants \$6.5 production weight by existing

E4 exit criterion

Criterion	Status
Every Group B package listed in Ch.24 has a section	Met
File-level purpose for bitnet/utis/training/portfolio/replay/analytics/expansion/regime/msip/multi_llm/retrieval/events/search/monitoring	Met
Explicit non-authority vs spine	Met
Deep train/BitNet tutorials	Out of scope (lineage audits + topics)

Next

- **E5** - Group C dormant (strategies, scanner, journal, cognitive, feedback, llm_research, data_ingestion, uat, ui)
- **E6** - Remaining scripts outside research/governance/analysis catalogs

Related: Ch.10 (../10-four-scoring-engines.md) * Ch.18 MSIP (../18-field-guide-governance.md) * Ch.23 (../23-multi-llm-coordination.md) * [E1](#) * [E3](#) * [Ch.24](#)

Encyclopedia E5 - Dormant / Archived Surfaces (Group C)

Phase: E5
Status: DONE (honest one-line + reopen rules - do not deep-read by default)
Date: 2026-08-07
Parent: [Chapter 24](#) * [Encyclopedia index](#)

Scope

Group C packages were **intentionally deprioritized** by the architecture book because they are not on the **active production candle->order story** - scaffold, multi-strategy experiments, sidecars with zero spine consumption, UAT harnesses, or **archived** parallel rails.

Package / tree	.py files	Default posture
src/strategies/	18	DORMANT scaffold (S01-S10)
src/scanner/	6	DORMANT / ORPHANED vs live single-candle spine
src/journal/	6	DORMANT identity/prove layers
src/cognitive/	2	DORMANT sidecar bus (F-012)
src/feedback/	2	DORMANT
src/llm_research/	5	DORMANT offline LLM research pipeline
src/data_ingestion/	6	MIXED - see §7 (not all dead)
src/uat/	4	DORMANT multi-strategy UAT harness

src/ui/	1	Empty / remnant package
archive/**	~19	ARCHIVED do not import from production
Total	~69	

Rule: DORMANT != delete. Means **do not reopen without explicit intent** and a reopen checklist.

How to use

1. **Default:** skip deep-reading these packages for spine work.
2. **If reopening:** read the package row + reopen conditions below; run construction protocol; do not silently wire into EngineRunner.
3. **Archive:** read only for forensics; never re-import as live path without a full design.

1. src/strategies/ - S01-S10 multi-strategy scaffold (18)

Posture: DORMANT - full strategy module framework exists; live production still runs the **single CRT spine**, not StrategyOrchestrator multi-strategy hot path (see F-013 class).

File	Purpose (one line)
base_strategy.py	ABC for S1-S10
strategy_result.py	Universal strategy output contract
strategy_intent.py	Typed hypothesis intent from strategies
intent_builder.py	StrategyResult -> intent adapter
strategy_orchestrator.py	Run all 10 strategies per candle + aggregate
strategy_package.py	Strategy package / registry architecture target
strategy_registry.py	Load/list StrategyPackage versions
s01_crt_wrapper.py	S1 CRT wrapper (closest to live spine)
s02_mean_reversion.py ... s10_trap_strategy.py	S2-S10 alternate strategies (MR, breakout, stat-arb, grid, scalp, news, ML ensemble, pattern, trap)
__init__.py	Package exports

Reopen if: multi-strategy production is an explicit goal + ValidationReport path + single-spine parity.

Do not: assume S08/S06 etc. are live because files exist.

Related: E3 multi_strategy_validator / strategy_backtest can exercise them offline.

2. `src/scanner/` - multi-symbol scan rail (6)

Posture: **DORMANT / ORPHANED** relative to live single-candle spine (scan->allocate->loop class, F-013).

File	Purpose
<code>scanner.py</code>	Multi-symbol scanner core
<code>universe.py</code>	Symbol universe definition
<code>ranker.py</code>	Rank scan hits
<code>signal_pool.py</code>	Pool of candidate signals
<code>spine_adapter.py</code>	Only designed path scanner -> <code>EngineRunner.run()</code>
<code>__init__.py</code>	Package marker

Reopen if: multi-symbol live allocation is authorized and wired end-to-end through Ultron.

Do not: confuse with `archive/inout_legacy/.../scanner.py` (different archived rail).

3. `src/journal/` - trade identity / provenance (6)

Posture: **DORMANT** as a production journal system; contracts exist for future correlation.

File	Purpose
<code>trade_identity_v1_0.py</code>	Minimal immutable trade identity (alert -> decision link)
<code>trade_provenance_v1_0.py</code>	Config/model/promotion lineage keyed by <code>trade_id</code>
<code>trade_execution_link_v1_0.py</code>	Broker/venue correlation layer (disposable Tier 0B)
<code>trade_logger.py</code> / <code>schema.py</code>	Journal logging + schema
<code>__init__.py</code>	Package marker

Note: Live logging often uses `src/utls/trade_logger.py` (E4) instead.

Archive twin: `archive/dead_code/journal/*` - older copies; ignore for new work.

4. `src/cognitive/` - cognitive bus (2)

Posture: **DORMANT sidecar** - async cognitive processing; **zero spine consumption** (F-012).

File	Purpose
<code>cognitive_bus.py</code>	Async cognitive processing bus
<code>__init__.py</code>	Package marker

Reopen if: explicit cognitive sidecar product goal; never block `EngineRunner` on bus health.

5. `src/feedback/` - AI feedback (2)

Posture: DORMANT.

File	Purpose
ai_feedback.py	AI feedback collection/loop (scaffold)
__init__.py	Package marker

Reopen if: closed-loop learning product is designed with measurement contract + no silent train-in-prod.

6. src/llm_research/ - offline LLM research pipeline (5)

Posture: DORMANT offline pipeline (pattern -> policy -> forward test). Not live LLM gate.

File	Purpose
pattern_extractor.py	Extract patterns via LLM offline
policy_builder.py	Build policies from extracted patterns
forward_tester.py	Forward-test generated policies
evaluator.py	Evaluate LLM research outputs
__init__.py	Pipeline architecture package

Distinct from: live llm_inference_client (E1, fail-open advisory) and multi-LLM ops (E4/Ch.23).

7. src/data_ingestion/ - MIXED (6)

Not pure dormant. Parts are legacy DB-oriented; parts are **LIVE integrity** used by research/backtest.

File	Posture	Purpose
dataset_integrity.py	LIVE-research	L3 whole-dataset sequence integrity preflight (F-039: limited call sites)
ohlcv_schema.py	LIVE-contract	Historical OHLCV column contract
session_autoderive.py	RESEARCH	Learn tradable session from data (vs hardcoded calendars)
xauusd_phase1_candidate.py	RESEARCH freeze	XAUUSD M15 Phase-1 candidate identity verification
historical_fetcher.py	LEGACY / convention tension	Multi-pair OHLCV into TimescaleDB (repo doctrine: no DB - prefer src/inout/* CSV fetchers)
__init__.py	-	Package marker

Prefer for new data I/O: src/inout/* fetchers (E1/Ch.15).

Prefer for integrity: dataset_integrity at backtest/research preflight when required.

8. src/uat/ - multi-strategy UAT harness (4)

Posture: **DORMANT** relative to single-spine production; useful only if multi-strategy returns.

File	Purpose
uat_runner.py	Orchestrates UAT areas for 10-strategy system
monte_carlo.py	Bootstrap resampling robustness checks
kill_switch.py	Daily/weekly loss kill-switch state machine
__init__.py	Package marker

Note: `kill_switch` ideas may overlap live risk concerns - production capital authority remains **UltronRiskGate** (E1/Ch.14).

9. `src/ui/` - remnant (1)

File	Purpose
__init__.py	Empty/remnant UI package

Archive: `archive/ui_legacy/dashboard_ARCHIVED_2026_05_02.py`, `archive/dead_code/ui/dashboard.py`.

Live UI surface: control-plane localhost dashboard (E1/Ch.22), not `src/ui/`.

10. `archive/**` - ARCHIVED (do not import)

Path	Why archived
<code>archive/inout_legacy/ARCHIVED_2026_05_02/</code>	Parallel execution rail (scanner/state_machine/controller/db) - bypassed spine; SQLite violated no-DB (Ch.15)
<code>archive/dead_code/**</code>	Extracted dead modules (old bitnet smoke, journal, ui, insight_reporter, bitnet_feature_builder)
<code>archive/ui_legacy/</code>	Old dashboard
<code>archive/scripts/</code>	Old historical data runner scripts (typo'd names preserved)
<code>archive/zips/</code>	Zip backups - not code authority

Replacement for archived INOUT rail: `HookedLiveEngine.process` (E1).

Replacement for data I/O: `src/inout/*` CSV fetchers.

Reopen checklist (any Group C package)

1. Explicit user/owner goal (not “file looks interesting”).
2. Classify against construction protocol / change contracts.
3. Prove callers (or add them) - no silent hot-path wire.
4. Measurement: if economic claim -> pre-registration + M4-class gate (E2).
5. If multi-strategy/scanner: address F-013 single-spine reality.
6. Never re-enable `archive/inout_legacy` as a second spine.
7. SESSION LOG + docs decision (Ch.17 drift protocol).

Danger flags

Flag	Meaning
DORMANT	Present; off active story; deprioritize
ORPHANED	Built path with no live caller (scanner/portfolio class)
ARCHIVED	Explicitly retired under archive/ - forensic only
MIXED	Package contains both live-adjacent and legacy files
CONVENTION_BREAK	e.g. TimescaleDB fetcher vs no-DB doctrine

E5 exit criterion

Criterion	Status
All Group C packages from Ch.24 have one-line map	Met
Archive trees listed with replacement path	Met
data_ingestion MIXED called out (not false dormant)	Met
Reopen checklist	Met
Full strategy design revival	Out of scope

Next

- **E6** - Remaining scripts outside research/governance/analysis catalogs (operator script matrix residual).

Related: Ch.15 INOUT ([../15-live-execution-and-inout.md](#)) * [E1 spine](#) * [E3 multi-strategy validators](#) * [Ch.24](#) * F-012 / F-013

Encyclopedia E6 - Remaining Operator Scripts (Group D residual)

Phase: E6

Status: DONE (category catalog for scripts not covered by E2/E3)

Date: 2026-08-07

Parent: [Chapter 24](#) * [Encyclopedia index](#)

Scope

Already catalogued	Where
scripts/research/ (126)	E2
scripts/governance/ (31)	E3
scripts/analysis/ (115)	E3
scripts/maintenance/ (9)	E3

E6 residual under scripts/	Count
data/	22
training/	13
root scripts/*.py	10
context/	5
misc/	5
evaluation/	3
export/	3
groq_bridge/	3
backtest/	2
control_plane/	1
metrics/	1
multi_llm/	1
portfolio/	1
E6 total	~70
All scripts/**/*.py	~351

Design rule: operator catalog by **job** (fetch data, train model, run backtest, multi-LLM context) - not 70 essays.
Prefer `src/` kernels; scripts stay thin.

Group: D (operational scripts).

Default danger: many **WRITE** data/, models/, results/, configs hashes.

How to use

Job	Directory
Fetch / convert / verify OHLCV	scripts/data/ -> src/inout/* (E1)
Train BitNet / RR / TradeNet / Gaussian	scripts/training/ -> src/training, src/bitnet (E4)
Portable Mind / multi-LLM context	scripts/context/ -> Ch.23
Control-plane server	scripts/control_plane/run_server.py -> Ch.22
Manual / debug backtest	scripts/backtest/ -> src/runtime/backtest_v2 (E1)
Groq bridge workflow	scripts/groq_bridge/
Model export / bootstrap	scripts/export/
One-off parity / zone / replay tools	scripts/misc/

Authoritative runnable catalog when regenerated: docs/reference/cli-matrix.md + docs/reference/script-matrix.md (E3 generators).

1. scripts/data/ - market data I/O (22)

Posture: LIVE ops for corpora. Writes under data/.

Fetch (thin wrappers over `src/inout`)

Script	Purpose
fetch_candles_mt5.py	MT5 -> CSV
fetch_candles_alphavantage.py	Alpha Vantage FX
fetch_candles_hummingbot.py	Hummingbot/exchange OHLCV
fetch_perp_funding.py	Binance perp funding/basis
fetch_crypto_ccxt.py	CCXT crypto fetch
fetch_forex_yfinance.py	Yahoo Finance forex
fetch_and_verify_binance.py / fetch_and_verify_mt5.py	Fetch + verify pipelines
check_availability.py / check_yfinance.py / test_yf_periods.py	Connectivity / period checks

Build / convert / verify

Script	Purpose
convert_binance_m1_to_m15.py / misc cousin resample	M1->M15 resample

convert_bnb_to_csv.py / csv_to_excel.py / split_bnb_excel.py	Format converters
build_m15_unified.py / unified_data_builder.py / prepare_data.py	Unified dataset builders
build_rr_dataset.py / build_tradenet_dataset.py	Train-set builders (label-sensitive)
generate_vectors.py	Feature vector generation helper
verify_output.py	Output verification

Prefer: production loaders + dataset_integrity (E5 MIXED) for sequence checks.

Danger: LABEL_SENSITIVE datasets (RR/TradeNet) - F-022 class.

2. scripts/training/ - model training CLIs (13)

Posture: OFFLINE. Artifacts under models/. No automatic production authority.

Script	Purpose
train_pipeline.py	Main train pipeline CLI
train_rr_model.py	RR pattern model train
train_bitnet.py / train_bitnet_contract_c.py	BitNet / CONTRACT-C train
train_trade_net_v2.py	TradeNet v2 train (unwired fusion - F-005)
train_gaussian_xauusd.py	Gaussian train path (artifact lineage caution - F-060)
phase5_calibration.py	Phase-5 calibration gate CLI
build_stage1_dataset.py	Stage-1 dataset for LLM/train
auto_tuner.py / auto_tuner_multi.py / auto_tuner_gemini_gate.py	Parameter tuner family
show_xauusd_gaussian_promotion.py	Promotion display helper
_write_hashes_closure_narrative.py	Hash/closure narrative helper

Related: E4 src/training/, E4 src/bitnet/, E3 promote path for any production registration.

3. Root scripts/*.py (10)

Script	Purpose	Danger
update_config_hash.py	Config hash helper (pair with maintenance _compute_hash)	HASH write
export_model_registry.py	Export model registry snapshot	read/write artifacts
auto_train_from_opportunities.py	Auto-train trigger from opportunities stream	TRAIN write
validate_integration.py	Integration validation harness	-

build_consolidated_docs.py	Docs consolidation builder	docs write
extract_folder_structure.py	Repo tree extract	-
rag_index.py	Build/update RAG index (src/retrieval)	optional deps
_gate5_compliance_pass.py	Gate-5 compliance counting helper	observe
tmp_cert_worktrees.py	Worktree certification temp helper	ops
__init__.py	Package marker	-

4. scripts/context/ - Portable Mind / multi-LLM (5)

Posture: OPS for multi-LLM workflow (Ch.23). Regenerates gitignored context/*.md.

Script	Purpose
build_context.py	Compile trigger - rebuild Portable Mind context/*.md
log_turn.py	Append multi-LLM turn to ledger
discussion.py	Discussion reconstruct / rewind
pack_story.py	Bounded story pack for handoff
seed_build_queue.py	Seed multi_llm/build_queue.jsonl

Related: E4 src/multi_llm/, root multi_llm/.

5. scripts/control_plane/ (1)

Script	Purpose
run_server.py	Start localhost control-plane HTTP server (:8787)

Related: E1 src/control_plane/, Ch.22. **No auth / localhost only.**

6. scripts/backtest/ (2)

Script	Purpose
manual_backtest.py	Manual backtest entry wrapper
backtest_debug_harness.py	Debug harness around backtest path

Related: E1 src/runtime/backtest_v2.py. Prefer official CLI entry points from cli-matrix when present.

7. scripts/export/ (3)

Script	Purpose
--------	---------

<code>export_bitnet_model.py</code>	Export BitNet model artifacts
<code>generate_bootstrap_model.py</code>	Bootstrap model generation
<code>regen_bitnet_35.py</code>	BitNet 3.5 regen helper

Danger: model artifact writes; no auto-promote.

8. scripts/evaluation/ (3)

Script	Purpose
<code>run_benchmark.py</code>	Benchmark runner
<code>report.py</code>	Evaluation report generation
<code>__init__.py</code>	Package marker

9. scripts/groq_bridge/ (3)

Script	Purpose
<code>prepare_retrospective.py</code>	Prepare retrospective pack for Groq/LLM
<code>ingest_response.py</code>	Ingest LLM response into workflow
<code>apply_llm_suggestions.py</code>	Apply suggestions (review carefully - may write)

Danger: treat apply step as human-gated.

10. scripts/misc/ (5)

Script	Purpose
<code>run_parity.py</code>	Parity run helper
<code>trade_replay_validator.py</code>	Trade replay validation
<code>build_zone_registry_from_trades.py</code>	Zone registry from trades (label-sensitive)
<code>resample_m1_to_m15.py</code>	Resample utility
<code>bitnet_ternary_inference.py</code>	BitNet ternary inference probe

11. Single-file operator CLIs

Script	Purpose
<code>metrics/extract_metrics.py</code>	Extract metrics from run artifacts
<code>multi_llm/initiate_plan.py</code>	Initiate multi-LLM plan cycle

portfolio/replay_allocator.py	Portfolio allocator shadow replay (F-013 class)
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Cross-links (do not duplicate)

Need	See instead of reinventing
Research qualify / phase drivers	E2
Promote / SITS / feature DAG	E3
Session log rotate / config hash	E3 maintenance
In-repo libraries behind scripts	E1 / E4

Danger flags (E6)

Flag	Meaning
WRITES_DATA	Touches data/** corpora
WRITES_MODELS	Touches models/**
WRITES_CONFIG_HASH	Production config hash paths
TRAIN_OFFLINE	No live trading authority
LOCALHOST_ONLY	Control plane / health
HUMAN_GATE	Groq apply / auto-train / promote-adjacent

E6 exit criterion

Criterion	Status
All scripts/ top-level dirs either in E2/E3 or listed here	Met
Residual ~70 scripts categorized by job	Met
Pointers to cli-matrix / script-matrix	Met
Per-script full API docs	Out of scope

Encyclopedia series status

Phase	Status
E0 Charter	DONE
E1 Spine Group A	DONE
E2 Research	DONE

E3 Governance	DONE
E4 Sidecars	DONE
E5 Dormant/Archive	DONE
E6 Remaining scripts	DONE

Optional later: E1b (src/features/ registry depth), per-program research deep-dives, or machine-generated JSONL twin of all encyclopedia rows.

Related: [E2](#) * [E3](#) * [E4](#) * Ch.22 (../22-control-plane.md) * Ch.23 (../23-multi-llm-coordination.md) * [Ch.24](#) * docs/reference/cli-matrix.md

Appendix A2 - Unresolved Questions (Rollup)

Status of this chapter: Written (updated Grok review pass, 2026-08-07)

Why this chapter exists

Per this book's own charter: "if understanding is impossible, do not invent." Every chapter that hit a genuine gap in evidence recorded it locally rather than guessing. This appendix collects those in one place - and, after the Grok review pass, also records what was **closed in the book** without changing production code.

What you need to already know

This is a reference appendix, not a narrative chapter - each item links back to the chapter that owns the full context.

Resolved in the book (Grok review pass - no code changes)

Item	Resolution	Chapter
INOUT <code>src/inout/</code> vs <code>archive/inout_legacy/</code>	Name collision. Current <code>src/inout/</code> = data fetchers only; archived tree = parallel execution rail retired 2026-05-02	Ch.15
Program 9 missing from research chapter	Chaptered from pre-registration (<code>docs/research/preregistration-program-9.md</code>); run verdicts still artifact-bound	Ch.19
ATR / F-057 "flag only" passivity	Proposed remediation options + trade-offs added; still OPEN pending owner	Ch.13
Measurement Contract gap	5-step seal path proposed; still 0 sealed instances in-repo unless registry says otherwise	Ch.20
MSIP "no treatment"	Orientation brief in governance field guide	Ch.18
Density / meta overwhelm	Quick Start chapter added	Ch.00
22 un-chaptered packages	Orientation map added (not full chapters)	Ch.04

Still open - book-scope gaps

- **Repository Encyclopedia** - charter in [Chapter 24](#);
E0-E6 + E1b + JSONL twin DONE (`docs/book/encyclopedia/`, including `encyclopedia_rows.jsonl`). Optional later: relationship graph edges in JSONL; agent query UI.
- **Full-depth Part VI-VIII** - chapters 16-23 remain orientation-level relative to their source corpora (~250 governance files, ~475 research files). Deepening is expected future book work.
- **Optional MSIP dedicated chapter** - brief exists; full gate-by-gate narrative only if MSIP re-enters active execution.
- **Program 9 numeric verdict** - design is chaptered; Stage-1/Stage-2 results must still be read from artifacts / findings, not assumed.

- **Program 7 (Open Interest)** - named as out-of-scope for Program 6; status not independently verified as run.
- **Deeper Part VIII implementation walk** - agent/control-plane/multi-LLM chapters improved slightly but are not full replacements for docs/reference/agent-reference.md.

Still open - real repository issues (code/doc; not fixed by this book pass)

User instruction for this pass: **no code changes**. These remain tracked, not remediated.

- docs/reference/schemas.md §4 still documents the 38-dim v3.0 feature tuple; live code is 39-dim v4.0. Textbook DOC_DRIFT. See [Chapter 7](#). *Recommended separate Doc Sync turn - not part of this PDF/book narrative pass.*
- docs/reference/testing.md file-count header ("116 files across 15 directories," dated 2026-06-12) is stale vs ~1,200+ files under tests/. See [Appendix A1](#).
- ATR unit mismatch in compute_crt_levels - real geometry defect; options A/B/C in [Chapter 13](#); awaiting owner decision.
- F-057 crtconfig programmatic split-brain - OPEN; options 1/2/3 in Chapter 13; "backtest Phase 2."
- Research Measurement Contract - schema frozen, 0 sealed MC-* (verify registry); path in [Chapter 20](#).

Verification gaps (assumed from docs; not re-proven this pass)

- Whether Wyckoff / Market Profile / Order Flow interpreters have code beyond scaffolding ([Chapter 9](#)).
- Whether every tool in docs/reference/agent-reference.md is end-to-end reachable ([Chapter 21](#)).
- Full body of docs/reference/architecture.md beyond header/orientation ([Chapter 4](#)).

Out of scope for this pass (explicit)

- Fixing schemas.md / testing.md primary docs (Doc Sync turn).
- Implementing ATR or F-057 remediations (code + parity + promotion).
- Sealing MC-* instances or running Program 9 Stage drivers.
- Adding CLAUDE.md companion-row (already present from an earlier session).

Previous: [Appendix A1 - Testing the System](#) * **Next:** - (last chapter; return to [the Table of Contents](#))

Related: every chapter linked above.

Memory: none dedicated - this appendix is the book's own open-loop tracker.